Launch and Initial Checkout of the Psyche Spacecraft for NASA's Discovery Program

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David Y Oh[*], Nora Alonge[†], Sarah H. Bairstow[‡], Abhijit Biswas[§], Carl Chesko[**], Dan M. Goebel[††], Travis Imken[‡‡], Jennifer Maxwell[§§], Arthur Na-Nakornpanom[***], Austin Nicholas[†††], John Steven Snyder[‡‡‡], and K. Charles Wang[§§§]
Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, 91109, USA

Ian Johnson[****], Paul Linggi[††††], Peter Lord[‡‡‡‡], Shane Malone[§§§§], and Zack Pirkl[*****]
Maxar Space Systems, Palo Alto, CA, 94303, USA

Jodie B. Ream[†††††], Benjamin P. Weiss[‡‡‡‡‡], and Rona Oran[§§§§§]
Massachusetts Institute of Technology, Department of Earth, Atmospheric and Planetary Sciences, Cambridge, MA, 02139

Jose M. G. Merayo[******]
DTU Space, Technical University of Denmark (DTU), Copenhagen, Denmark

On October 13, 2023, the Psyche spacecraft was launched into space from Kennedy Space Center. This paper provides an overview of the Psyche project and its progression through launch and its first six month of operations in space. It describes preparations for launch, initial testing, and operations of the spacecraft, including in-space tests of the spacecraft's unique power and electric propulsion system, validation of its all-electric 3-axis momentum management strategy, and demonstration of the Deep Space Optical Communications technology payload. Psyche is the first mission to use Hall thrusters for interplanetary travel and carries NASA's first optical communications demonstration from Mars distances. It has successfully completed its initial checkout phase and transitioned to cruise operations as it starts its journey to (16) Psyche.

I. Introduction

On October 13, 2023, the Psyche spacecraft was launched into space from Kennedy Space Center on a Falcon Heavy Rocket. The launch was the beginning of the spacecraft's 2.2-billion-mile journey to the asteroid (16) Psyche, located between the orbits of Mars and Jupiter in the outer part of the main asteroid belt. *Psyche: Journey to a Metal*

[*] Chief Engineer for Operations, Psyche Project, 4800 Oak Grove Drive.
[†] Science and Instrument Operations Team Lead, Psyche Project.
[‡] Mission Planner, Psyche Project.
[§] DSOC Project Technologist.
[**] Power Systems Engineering Lead, Psyche Project.
[††] Spacecraft Chief Engineer, Psyche Project.
[‡‡] Psyche Launch Phase Lead, Psyche Project.
[§§] Project Systems Engineer, Psyche Project.
[***] Senior Thermal Engineer and Mission Operations Thermal Lead, Psyche Project.
[†††] Spacecraft Guidance, Navigation and Controls Systems Engineer, Psyche Project.
[‡‡‡] Principal Engineer, Electric Propulsion Group.
[§§§] Launch Systems Engineer, Psyche Project.
[****] Principal Propulsion Systems Engineer, 4875 Fabian Way.
[††††] Senior Thermal Engineer, Thermal Engineering.
[‡‡‡‡] Psyche Maxar Solar Electric Propulsion Chassis Project Manager.
[§§§§] Systems Engineering Manager, Psyche Solar Electric Propulsion Chassis.
[*****] Power Systems Engineer, Psyche Solar Electric Propulsion Chassis.
[†††††] Magnetometer Instrument Scientist.
[‡‡‡‡‡] Psyche Deputy Principal Investigator and Magnetometry Investigation Lead.
[§§§§§] Magnetometry Investigation Scientist.
[******] Co-Investigator and Magnetometer Lead Engineer

World is the 14th mission in NASA's Discovery program. It is led by principal investigator Dr. Lindy Elkins-Tanton of Arizona State University, is enabled by solar electric propulsion, and will use SPT-140 Hall effect thrusters to rendezvous and orbit the largest metal-rich asteroid in the solar system. The spacecraft uses no chemical propulsion and is the first mission to operate Hall thrusters in interplanetary space. In addition to its science instruments, it carries a Deep Space Optical Communications (DSOC) technology demonstration payload, which is NASA's first optical communications demonstration from Mars distances.

Previous work has described both the early concept design for the Psyche mission and the evolution of the technical design through its Systems Integration Review and environmental test campaign.[1,2] This paper provides an overview of the Psyche spacecraft's progression through launch and its first six month of operations in space. It includes a brief description of the project's scientific objective and an overview of the mission architecture and the spacecraft's overall design. It then describes launch preparations and presents results from launch and initial testing and operations of the spacecraft in space, including tests of the spacecraft's unique power and electric propulsion system and demonstration of the operational strategy to manage the spacecraft's 3-axis momentum during interplanetary cruise using only electric propulsion. The spacecraft has successfully completed its initial checkout phase and has transitioned to cruise operations as it starts its 5.9 year journey to Psyche.

II. Mission Overview

A. Science Objectives and Payloads

The state of knowledge about (16) Psyche and fundamental objectives of this mission remain the same as described in previous work.[2] Based on recent measurements,[3] Psyche* is likely to be primarily made of metal, based upon its high radar albedo,[4] high thermal inertia,[5] featureless reflectance spectra,[6,7] and relatively high density.[8] Work with the Atacama Large Millimeter/submillimeter Array (ALMA) measuring Psyche's thermal emissions has concluded that Psyche likely has a fine-grained metallic surface, with grains mainly less than 1 mm in diameter.[9] Together these attributes, along with its large size,[10] make the asteroid unique in our solar system.

The primary hypothesis for the formation of Psyche continues to be that it is part of the metal core of a planetesimal (a small body formed within about one million years of the first solids in our solar system) that had its rocky exterior largely, but not entirely, removed by collisions with other solar system bodies. Though this is the primary formation hypothesis, it is far from sure: We are certain to be scientifically surprised at what we discover when we arrive at Psyche.

The Psyche mission has three broad scientific goals: 1) Understand a previously unexplored building block of planet formation: iron cores, 2) Look inside the terrestrial planets, including Earth, by directly examining the interior of a differentiated body, which otherwise could not be seen, and 3) Explore a new type of world. For the first time, examine a world made not of rock or ice, but of metal.

The mission's science objectives as derived from these goals are to A)Determine whether Psyche is a core, or if it is primordial unmelted material, B) Determine the relative ages of regions of Psyche's surface, C) Determine whether small metal bodies incorporate the same light elements into the metal phase as are expected in the Earth's high-pressure core, D) Determine whether Psyche was formed under conditions more oxidizing or more reducing than Earth's core and E) Characterize Psyche's topography.

One of the instruments that the spacecraft is flying is the Magnetometer, which consists of two three-axis fluxgate magnetometers placed in a gradiometer configuration on a boom, led by MIT and delivered by the Technical University of Denmark[11]; two multispectral imagers led by Arizona State University and delivered by Malin Space Science Systems; and a gamma-ray and neutron spectrometer (GRNS) led and delivered by John Hopkins Applied Physics Laboratory. The mission will also run an important gravity investigation led by MIT and facilitated by processing X-band radio transmissions through the spacecraft's telecommunications system. Additional information on the instrument suite, including a summary of results from initial checkout testing, can be found in section V.F and reference 12.

An early-forming planetesimal would be expected to heat significantly from radiogenic ^{26}Al, and would today be completely cold. If the Magnetometer detects a large magnetic field at Psyche, then Psyche is almost certainly a planetesimal core: its parent body had a core magnetic dynamo and the core's colder exterior permanently recorded the magnetic field.[11] Further descriptions of the scientific aspects of the mission can be found in references 3 and 8.

In addition to the science instruments, Psyche also includes a technology demonstration payload: a Deep Space Optical Communications (DSOC) system that conducts high-rate communications using lasers between the spacecraft and ground stations on Earth in support of future human and robotic deep space missions. The DSOC system was developed and implemented to conduct NASA's first optical communications demonstration from Mars distances, covering 0.4 – 2.7AU Earth-Probe distance.[13] The objective is to validate space flight and ground technologies for implementing photon-efficient signaling while demonstrating at least an order of magnitude higher data-rates than state-

* The IAU designation is (16) Psyche. For clarity, we will use Psyche throughout this paper

of-art deep space telecommunication systems. Further description of the DSOC system can be found in section V.G.

B. Mission Architecture

The Psyche mission includes the five mission phases shown in Figure 1, each of which is distinguished by different operational goals and strategies: Launch, Initial Checkout, Cruise, Approach, and Orbital Operations.

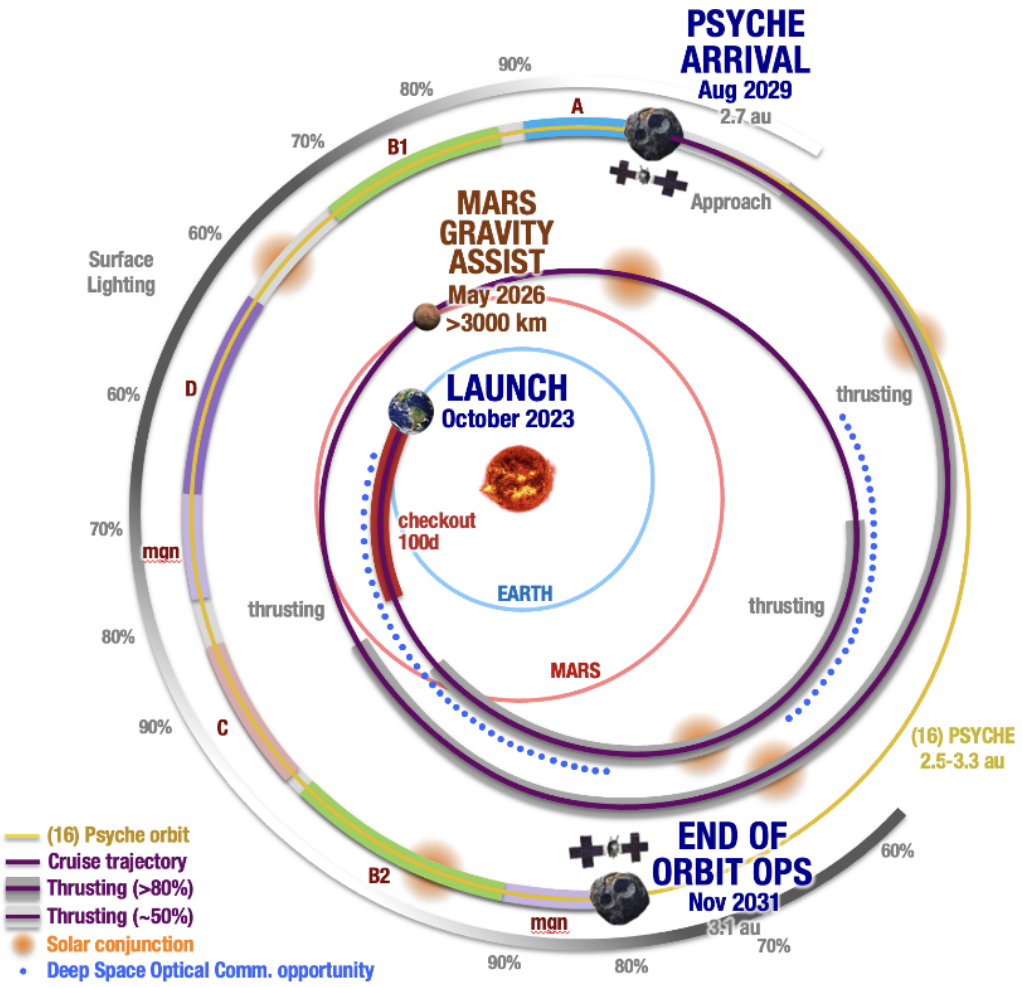


Figure 1. Psyche mission overview.

Psyche launched aboard a SpaceX Falcon Heavy launch vehicle, and autonomously deployed its solar arrays and entered into a power-positive, thermally safe, stable attitude in two-way communications with the Earth. This was followed by a 103-day Initial Checkout (ICO) phase with the primary objective of assessing the health and performance of each subsystem to the level necessary to begin continuous thrusting. Some planned checkout activities and other new checkout activities were delayed until early in the Cruise phase. Further details of the Launch and Initial Checkout phases as flown are described in sections IV and V.

The Cruise phase is a 5.9-year period in which electric propulsion thrusting places the spacecraft on a trajectory to rendezvous with Psyche, with help from a Mars gravity assist in May 2026. During thrusting periods, one SPT-140 thruster is used for primary (mission) thrusting, interspersed with momentum management activities conducted using a separate SPT-140 thruster to offload momentum accumulated on the reaction wheels. Mission thrusting averages 80-90% duty cycle in the periods of Cruise during which there is active thrusting. The Cruise phase also includes substantial periods of optimal coast, during which mission thrusting ceases because it not necessary. The Δv delivered by the SPT-140 thrusters for primary propulsion during Cruise is approximately 5.8 km/s.

During Cruise, thrusting and other spacecraft operations are designed in 3- to 5-week segments consisting of command sequences uplinked to the spacecraft 1-2 weeks before the start of execution. Development of these sequences is a multiple-week process which typically begins with identification and characterization of activities which are

incompatible with thrusting, such as weekly Deep Space Network (DSN) communications tracks, DSOC demonstration opportunities, and engineering and momentum management activities.[14] The operations team generates a timeline of these "forced coast" activities, specifying times the spacecraft is allowed to thrust, from which the mission design team re-optimizes the Cruise trajectory and creates a detailed thrust design for the planning period. The thrust sequence and other sequence(s) are developed in parallel and then modeled together to ensure that there are no commanding conflicts and that the ensuing sequence of events will achieve the objectives of the planning period.

The 100-day Approach phase following Cruise completes capture into Psyche's sphere of influence and insertion into the initial orbit for science operations. Approach incorporates a campaign of optical navigation, with imaging sessions every few days enabling asteroid-relative navigation into the initial science orbit of the Orbital Operations phase.

Orbital Operations consists of a series of science orbits interspersed with electric propulsion transfers between orbits. The science orbits are labeled alphabetically from highest to lowest altitude (A, B1/B2, C, D) but are visited in a different order (A, B1, D, C, B2) derived from geometric constraints on science data collection. Over the course of 26 months, the mission will collect global magnetic field datasets at multiple altitudes, Doppler and ranging datasets contributing to a refined gravity model, elemental composition data over a large portion of the surface, and global imaging maps in both clear and color filters at different resolutions, plus a limited number of highest resolution images in the lowest altitude orbit. End of mission occurs on November 1, 2031.

The mission plan evolved in the year and a half prior to launch in response to launch delays from August 2022 to September/October 2022 and finally to October 2023. Although the mission timeline changed substantially, the principles behind the mission architecture remained largely unchanged and the updated mission addresses all the project's original science objectives. The full mission duration increased from 5 years to 8 years and the cruise duration increased from 3 years to 5 years, reducing time pressure to start thrusting before the Mars Gravity Assist and creating much longer optimal coast periods both before and after Mars. The ordering of science orbits was updated to reflect the different orbital geometry. This changed the timeline from a simple incremental step-down through the orbits from highest to lowest altitude (i.e. A, B, C, D) into a more complex ordering in which Orbit B is visited twice (i.e. A, B1, D, C, B2). This in turn generated some benefits for the science observing plan, including the elimination of eclipses in the lowest orbit and additional opportunities for instrument calibration. Although this change increased the number of electric propulsion orbit transfers from three to four, individual transfer times are shorter because Psyche is at a closer Sun range at that part of its orbit and therefore more solar power is available for electric propulsion. The current mission plan, along with a comparison to the previous plan for a 2022 launch, is described in more detail in Ref. 15.

C. Spacecraft Description

The Psyche spacecraft is based on a Solar Electric Propulsion (SEP) "Chassis" manufactured by Maxar in their 1300-series spacecraft product line. The power, electric propulsion, attitude control system and thermal-control subsystems in the Maxar-provided bus have extensive heritage from their Earth-orbiting commercial satellites. The high-power electric propulsion system provided by Maxar is critically important for performing the Psyche mission. Maxar has built over 40 Earth orbiting spacecraft that have accumulated over 250,000 hours of active electric propulsion operation in space. Over a dozen of these Maxar satellites have more than 20 kW of electrical power capability. JPL provided the spacecraft's main computer, flight software, and autonomous fault-protection systems to support for deep space operation. JPL also provided an X-band telecommunications systems (with a High Gain Antenna (HGA) provided by Maxar), and a low voltage power distribution assembly to complete the spacecraft bus. The flight software was developed in partnership with the Europa Clipper project from a new multi-mission JPL Flight Software Core Product Line. The Command and Data Handling System (C&DH) was a joint effort, with JPL and Maxar both providing components for Psyche.

The Psyche spacecraft with deployed solar arrays is shown in Figure 2 and the configuration without the stowed solar arrays mounted is seen in Figure 3. It features a rectangular, panel-box construction supported by an internal core graphite composite central-cylinder. The 2-m diameter HGA is hard mounted on the top (+Z) deck and canted at a slight angle to avoid the two instruments mounted on the graphite-composite booms. The DSOC technology demonstration experiment is seen mounted on the -X side and is co-bore sighted with the HGA to enable simultaneous communication during DSOC operation. The ±X closeout panels are removed in this figure to show the internal central thrust cylinder that houses the seven 82-liter xenon propulsion tanks that hold up to 1085 kg of propellant and support three 82-liter composite-overwrapped tanks outside the cylinder that hold 45.9 kg of nitrogen for the cold gas propulsion system.

Primary propulsion for the Psyche spacecraft is provided by four SPT-140 Hall effect thrusters, which were first flown by Maxar in 2018 and have been successfully operated on eight Maxar spacecraft to date. The 4.5 kW SPT-140 thrusters provide higher thrust and longer lifetime than the 1.35 kW SPT-100 system that has been successfully used on over 30 Maxar spacecraft since 2004. First operation of the electric propulsion system is discussed in section V.B. The design and operation of the cold gas propulsion system during spacecraft checkout is described in section IV.B. The

power system architecture is split between Maxar and JPL, with most of the hardware supplied by Maxar using elements of its EPS 2.0 power subsystem. The power subsystem is discussed in more detail in section V.C.

The Psyche spacecraft is three-axis stabilized using commercially available star trackers and ring laser gyros for inertial attitude and rate measurements. Attitude control is provided by four Honeywell reaction wheels (RWAs), which are mounted in a pyramid configuration inside the bus below the propellant tanks. Several sun sensors mounted on the ±Y panels are used to maximize the field of view coverage to detect the sun's location during safe mode. A nitrogen cold gas system was used to null the spacecraft rotation after tip-off from the launch vehicle, for initial RWA unloading before the electric propulsion system was enabled, and for a single safe mode entry during ICO. It is planned that all momentum unloading through the cruise phase will be performed using the electric propulsion system, and the cold-gas system will only be used going forward during safe mode in situations where the reaction wheels cannot be used. The operation of the attitude control system for Psyche is described in detail in Section V.D.

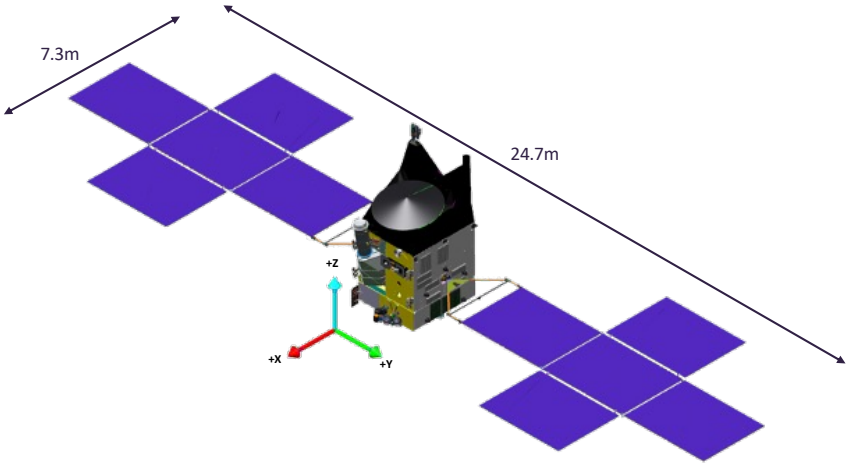


Figure 2. Psyche spacecraft with the large 5-panel solar arrays deployed.

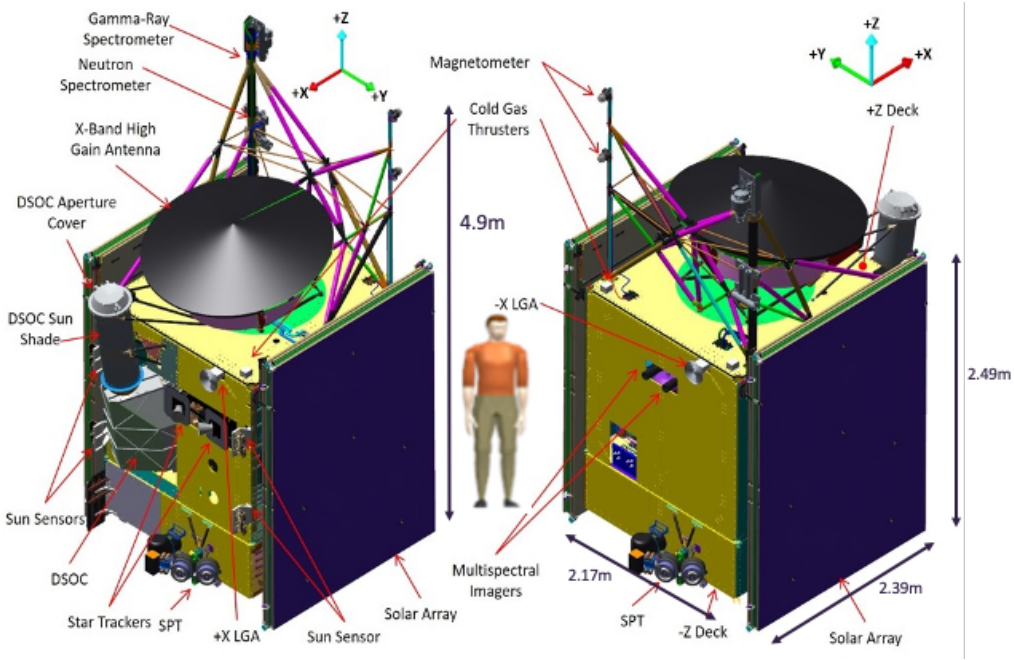


Figure 3. Psyche spacecraft without the solar arrays or thermal blanketing showing the various components installed on the bus.

A photograph of the spacecraft during assembly in ATLO with the +X side close out panel removed is shown in Figure 4. DSOC is on the left and two of the SPT-140 thrusters (with their red protective covers) on a stowed Deployable SPT Module (DSM) boom are clearly visible. The composite graphite boom structure mounted on the top +Z deck

features an optimized configuration design to support the instruments during launch vibrations using minimal structure mass. The booms, the associated support structure, and the HGA are completely covered in thermal blanketing to help maintain their temperatures within an acceptable range at the asteroid.



Figure 4. Psyche spacecraft during assembly.
DSOC on the left, SPT Thrusters at the Bottom (with red protective covers)

Psyche utilizes a fully redundant C&DH system with components developed by both JPL and Maxar. The Rad750 processor in the Psyche Compute Element (PCE) hosts the flight software that controls the spacecraft. As illustrated in Figure 5, the PCE controls the SEP chassis via two RS-485 serial data lines that use Maxar's proprietary M-Bus protocol to connect to the Maxar provided Attitude Control Electronics (ACE), the RS485 router, and the Smart Battery Tray that interfaces with the Li-Ion battery. The ACE in turn provides command and telemetry interfaces for ACS sensors and actuators. The 1553 bus provides the interface to the Miniature Inertial Measurement Units (MIMUs) and the two star trackers. Maxar's proprietary serial bus protocol is used internal to the SEP Chassis to interface to thermal, propulsion, pyro and power components.

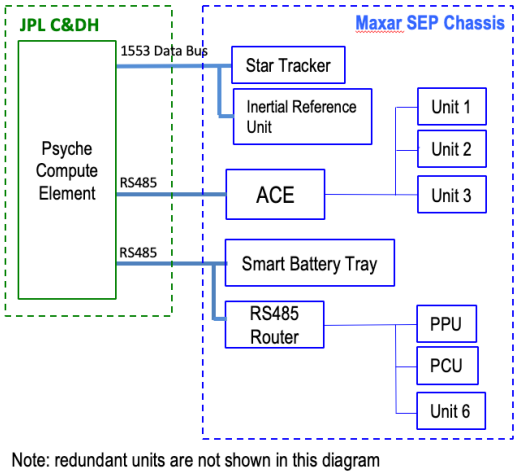


Figure 5. The Psyche Compute Element (PCE) controls the SEP Chassis through 1553 and RS-485 digital interfaces to Maxar hardware.

The Psyche spacecraft uses an X-band Small Deep Space Transponder (SDST) to communicate with NASA's Deep Space Network. The system uses a 100 W Traveling Wave Tube Amplifier (TWTA) to provide downlink to the DSN at a maximum data rates between 360 kbps and 150 kbps depending on the range from Earth when using the fixed HGA mounted on the +Z deck. Uplink at rates of up to 2 kbps are also supported using the HGA, or via one of three body-mounted Low Gain Antennas (LGAs) which are oriented to provide wide field of view uplink coverage. The spacecraft can also downlink via a selected LGA, which ensures a downlink data rate of at least 10 bps in safe-mode to the DSN at over 4 AU Earth-Probe distance. The graphite composite HGA, provided by Maxar, is based on their substantial experience manufacturing antennas for communications satellites. The remainder of the telecommunications equipment is provided by JPL and draws heritage from previous missions such as Juno and Dawn.

The thermal system architecture is similar to Maxar's geostationary orbit (GEO) satellites and uses heaters to maintain minimum allowable flight temperatures and passive radiators with heat pipes to dissipate excess heat. The unique addition is louvers installed over radiator surfaces to maintain the spacecraft temperatures in deep space over Psyche's sun ranges (1 AU to 3.33 AU Sun-Probe distance). The location of optical surfaces and louvres on the exterior of the panels is optimized to achieve allowable flight limit temperatures for the units installed on the interior of the panels. Figure 6 shows the spacecraft on its side in ATLO with the louvers seen as aluminum-color covers over the radiators on the left. An interesting issue with using louvers on the Maxar bus specifically on the Psyche mission was that the stowed solar arrays had to be mounted several centimeters farther off the Y-panels to clear the louvers. DSOC is located in a dedicated thermal cold zone separated from the rest of the heat pipe matrix to maintain DSOC's temperature-sensitive hardware below its maximum operating temperature. Figure 6 also shows two of the science instruments, the GRNS and Magnetometer, installed on the booms prior to full blanketing for thermal protection.



Figure 6. Psyche spacecraft photograph showing aluminum louvers (on the left) covering thermal radiators placed over the internally mounted high dissipation components. Also visible are the HGA and boom structure holding the GRNS and MAG instruments.

The fully assembled Psyche spacecraft with the two stowed solar arrays is shown in Figure 7. The spacecraft is mounted on the vibration table in preparation for the environmental testing that included sine-vibe, shock and acoustics. The spacecraft successfully completed both environmental testing and thermal-vacuum testing at the JPL facilities in Pasadena. The spacecraft was then shipped to Kennedy Space Center (KSC) for functional testing and reinstallation of the solar arrays, which were removed and shipped separately. Both the xenon and nitrogen propellants were loaded onto the spacecraft at launch base prior to launch.



Figure 7. Psyche spacecraft fully assembled on the vibe table with DSOC (with white sun-shade), the stowed solar arrays, and two SPT-140 Hall Thrusters installed on the DSM. The HGA is seen on top and an LGA and Two star-trackers are on the front.

III. Launch Vehicle Integration and Operations

Launch services for the Psyche mission were provided through the NASA Launch Services Program office (LSP) at KSC. In February 2020, a contract was awarded to Space Exploration Technologies Corporation (SpaceX) for a Falcon Heavy launch vehicle with a 30-month mission integration period before launch, originally slated for August 2022. At the time, the planned launch included two smaller NASA missions – EscaPADE and Janus – as secondary payloads mounted on a rideshare dispenser below the Psyche spacecraft. However, due in part to incompatibilities with Psyche's interplanetary trajectory, both EscaPADE and Janus were later de-manifested. The final launch configuration nevertheless retained the unpopulated rideshare dispenser because launch vehicle mission analysis (e.g., performance, coupled loads, harness design, etc.) had already assumed the presence of the dispenser, see Figure 8.



Figure 8. Psyche mated to SpaceX payload adapter, rideshare dispenser, and payload attach fitting.

The Falcon Heavy is a two-stage, partially reusable, heavy-lift launch vehicle. The first stage consists of a center core booster and two side boosters, together comprising 27 Merlin 1D engines that run on super-cooled liquid oxygen (LOX) and kerosene (RP-1) as propellants. The second stage is powered by a single Merlin 1D vacuum-optimized engine that is ignited shortly after stage separation from the first-stage center core. To achieve the hyperbolic injection energy needed for Psyche's trajectory, the Falcon Heavy flew in an "RXR" configuration in which the side boosters were returned to the launch site and the center core booster was expended. The payload fairing was equipped with standard hardware for ocean recovery.

Requirements associated with launch vehicle integration and operations were specified in the Psyche Launch Vehicle Interface Control Document (ICD), which included requirements for the launch targets, loads and dynamic environments, thermal limits, payload fairing close clearances, electrical interfaces, radiation environments, and contamination controls. The launch period was selected to be 21 days, balancing launch vehicle performance with planetary alignment and spacecraft cruise capabilities. Several mission-unique features were incorporated into the ICD, including a vibration isolation system to address high dynamic environments, Earth eclipse duration constraints to satisfy spacecraft thermal limits and to facilitate post-separation solar array deployment, a Transient Voltage Suppression system to mitigate lightning strikes while on the launch pad, and a gaseous nitrogen purge for the DSOC instrument to meet optical cleanliness requirements.

Mission integration was coordinated across JPL, Maxar, LSP, and SpaceX through biweekly remote and quarterly in-person meetings. A joint "fit check" activity at JPL in early 2022 verified the spacecraft's mechanical and electrical interfaces to the launch vehicle. The spacecraft was shipped via a US Air Force C-17 transport to KSC in late April 2022 for launch processing at the KSC Payload Hazardous Spacecraft Facility (PHSF). In late June 2022 the launch was delayed to October 2023 due to issues related to software testing. As a result, Psyche was moved out of the PHSF to an alternate payload processing facility at Astrotech Space Operations in nearby Titusville, Florida, where it spent several months in a quiescent "maintenance mode". Active launch processing resumed in June, and final loading of xenon and nitrogen propellant was completed in August 2023. SpaceX performed a full wet-dress rehearsal of the launch vehicle without the spacecraft, followed by a 7-second static fire test in late September as a demonstration of launch readiness.

Concurrently, spacecraft mate and fairing encapsulation were completed in anticipation of the October 5th opening of the launch period (see Figure 9).

In addition to integrating the spacecraft with the launch vehicle, several engineering milestones needed to be met to prepare for launch and early cruise. These included a series of reviews to demonstrate the project's readiness for launch and early cruise operations, ensuring that JPL's institutional engineering standards and requirements were completed, and gaining approval from both JPL and NASA management regarding the residual risks which would be present for the mission on launch day.

To demonstrate readiness for launch, the project and operations teams presented their plans for products, processes, and staffing for early operations, as well as ensuring that the teams were adequately trained to operate the mission. This proved challenging as part of the team supporting the delivery of procedures and products to operate the vehicle after launch were also focused on finalizing the pre-launch Verification and Validation (V&V) of the flight software, which was completed only a few weeks prior to launch. Operations training for this portion of the team occurred on a compressed "just-in-time" schedule, and the overall team was able to leverage their understanding of the vehicle's behavior built up during the V&V campaign. However, certain elements, such as having procedures, sequence products, and contingency plans pre-built, tested, and approved, were not fully completed pre-launch. Consequently, the plans for initial launch and operations activities were modified to fully support the operations team with individuals who had worked to build and test the spacecraft, and time was allocated in the schedule to complete these products post-launch.

Another significant pre-launch milestone was the internal Certification of Flight Readiness (CoFR) conducted by JPL. In this process, senior leaders from engineering directorates across the lab audited all design documents, review products, requirements closure, and delivery materials to ensure compliance with institutional standards for engineering processes. While preparing for this review requires significant effort, it serves as an effective mechanism to ensure that all open issues are resolved prior to launch. A key component of this process was the residual risk review, where project risks, waivers to institutional design principles, and significant anomalies were discussed. Once institutional leaders agreed that the mission was ready for launch, the project then discussed open technical and mission risks with the NASA program office to ensure that they fully understood this and other residual risks to the mission.

The most significant residual risks associated with the electric propulsion system were related to the uncertainty around how performance and life tests conducted on SPT-140 thrusters on the ground would reflect the actual operation of the thrusters in space. Considerable effort was taken to address this risk prior to launch. These efforts included: 1) testing of the qualification model SPT-140 thruster at both JPL and Glenn Research Center over a range of throttle conditions and facility pressures to examine the performance impacts of facility pressures compared to the lower pressure which would be encountered in space,[21] 2) computational life modeling performed at JPL to demonstrate that the life of the thruster over the Psyche trajectory at space-pressure conditions would be better than those that shown in SPT-140 life testing on the ground,[16,17] and 3) comparisons of predicted performance with the performance achieved during electric orbit raising performed on commercial satellites in Earth orbit while operating at high throttle levels (above 3 kW discharge power).[18] Despite these efforts, the full uncertainty could not be fully addressed prior to launch because the thrusters had yet to be operated across their full throttle range in space. As a result, mandatory in-space checkouts of the thrusters were incorporated into the post-launch initial checkout plan to measure their performance as early as practical to allow the operations team an opportunity to validate and/or correct thrust the throttle curves used for trajectory planning to account for differences in performance. With this mitigation in place, the residual risk was deemed acceptable for launch.



Figure 9. Psyche encapsulated in SpaceX payload fairing.

After a 7-day delay to complete verification of parameters used to control the cold gas thrusters, and an additional 1-day weather delay due to high winds, Psyche was successfully launched at 10:19:43 AM EDT on Friday Oct. 13, 2023 (Figure 10). Spacecraft separation occurred 62 minutes after liftoff with initial acquisition six minutes later via the DSN station at Canberra, Australia. Figure 11 shows the ascent timeline from the October 13 launch.



Figure 10. Psyche launch from pad LC-39A at Kennedy Space Center

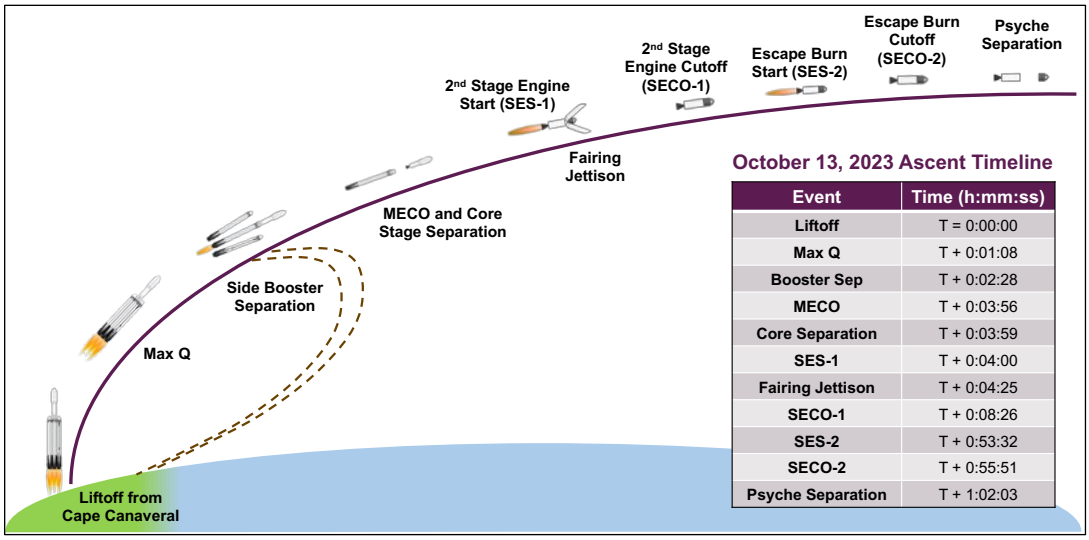


Figure 11. Psyche launch profile and separation timeline.

Parameter (units)	Target	Achieved	Diff (A-T)	Requirement
C_3 (km²/sec²)	34.0000	34.0023	0.0023	±0.25
DLA (deg)	28.0089	28.0153	0.0064	±0.25
RLA (deg)	113.7047	113.7071	0.0024	±0.50
X_{SC} (deg/sec)	0.00	-0.0979	-0.0979	±0.75
Y_{SC} (deg/sec)	0.00	0.2613	0.2613	±0.75
Z_{SC} (deg/sec)	0.00	0.0555	0.0555	±0.50

Table 1. Orbit injection performance

Post-launch orbit determination showed that the Falcon Heavy achieved excellent orbit injection performance, with hyperbolic targets (C_3, DLA, RLA; defined at 15 minutes after SECO-2) and spacecraft tip-off rates (X_{SC}, Y_{SC}, Z_{SC}; defined at Psyche separation) well within the ICD requirements, as shown in Table 1.

IV. Spacecraft Launch, Solar Array Deployment, and Initial Acquisition

A. Autonomous Launch Behavior

For the spacecraft, Psyche's Launch Phase began when the vehicle was powered on for the last time on the launch pad during countdown and concluded when the spacecraft received its first set of ground commands in space. During this phase, the spacecraft autonomously detected separation from the launch vehicle, nulled its rotation rate, deployed both its solar array wings, located the Sun, and entered a power-positive, thermally safe, and communicative safe-mode rotisserie spin.

On launch day, approximately an hour before liftoff, the ground team verified the final spacecraft configuration and commanded the spacecraft's onboard software to enter Launch Mode. In this state, the spacecraft actively monitors the state of four breakwires that connects the spacecraft to the launch vehicle. By design, separation is declared when the spacecraft detects that at least 3 of 4 break wires show a broken (separated) state for three seconds. The ground telemetry stream was disabled in the final minutes before liftoff.

After detecting separation, the spacecraft waited 30 seconds to drift away from the launch vehicle, powered on the MIMU, and then transitioned to Solar Array deployment Mode. The communications subsystem then began a four-minute warmup of the TWTA to enable downlink transmission.

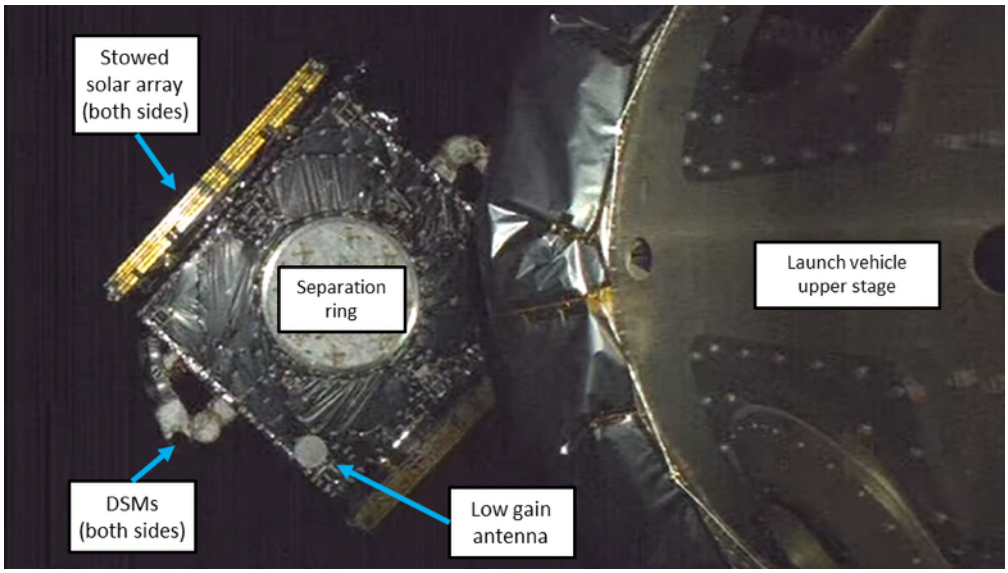


Figure 12. Separation as viewed from the SpaceX upper stage.

The transition to deployment mode configured the avionics and the guidance, navigation, and control systems for release of the solar arrays and initialized the on board autonomous solar array deployment behavior. This behavior, hard coded into flight software, exercised all primary and redundant release paths to ensure a successful array deployment. Entry into the behavior initiated a detumble from launch tip-off rates using the cold gas thrusters. To maximize the chance that ground operators would be able to observe the array deployment in real time, the spacecraft waited for TWTA warmup to complete prior to proceeding with the first solar array wing release. The +Y solar array wing (stowed on the spacecraft +Y body panel) was released first, followed by another optional spacecraft attitude rate null (ultimately not necessary), and then by release of the -Y wing. The behavior enforced a six-minute delay after each wing release for the array panels to unfurl, lock, and allow deployment dynamics to settle. To maximize the likelihood of a successful deployments, the deployment behavior "fired" each wing's eight release mechanisms four times, with only one successful "fire" required to release each mechanism.

The last action of the behavior was to initiate the spacecraft's transition into Cold Gas Safe Mode. The vehicle first actuated its solar array drive mechanisms to confirm the rotational position of each wing and then, since the deployed orientation of each solar array faces away from the sun line, the spacecraft rotated each wing to face the +X direction. The vehicle then began to search for the Sun by spinning slowly about the Y-axis. The separation attitude and tip-off rates delivered by the launch vehicle were close to ideal, so the sun was resolved by the sun sensors nearly immediately. The safe mode entry completed when the spacecraft entered a one revolution per hour rotisserie about the X-axis, with the +X axis pointed at the Sun. This rotisserie, combined with the near 90-degree Sun-Probe-Earth (SPE) angle, guaranteed that the -Z LGA field of view would periodically sweep past the Earth. Because of the rotisserie, spacecraft telemetry temporarily disappeared for about half of the first rotation. In the subsequent rotations, the ground team confirmed spacecraft commandability, stopped the rotation with the LGA pointed at Earth, and played back the recorded launch data. Figure 12 shows the vehicle at separation and Table 2 shows launch day post-separation timeline.

SA Deployment Milestone	Time from Separation (S+ H:MM:SS)
Separation	S = 0:00:00 - UTC 15:21:47
SA Deployment Behavior Start	S + 0:01:42
Detumble	S + 0:01:44 to 0:02:33 [duration: 0:00:49]
TWTA Warmup Complete	S + 0:05:33
Plus Y Array Deployment	S + 0:05:44 to 0:13:30 [duration: 0:07:46]
Minus Y Array Deployment	S + 0:13:31 to 0:21:17 [duration: 0:07:46]
Cold Gas Safe Mode Entry	S + 0:21:17
Solar Array Position Reacquisition	S + 0:22:38 to 0:34:42 [duration: 0:12:04]
Sun Search	S + 0:36:28 to 0:36:40 [duration: 0:00:12]
Sun Point	S + 0:36:41 to 0:42:15 [duration: 0:05:34]
Rotisserie Rate Achieved	S + 0:45:46
Rotisserie Stopped (Ground commanded)	S + 2:38:51

Table 2. Psyche Post-Separation Timeline

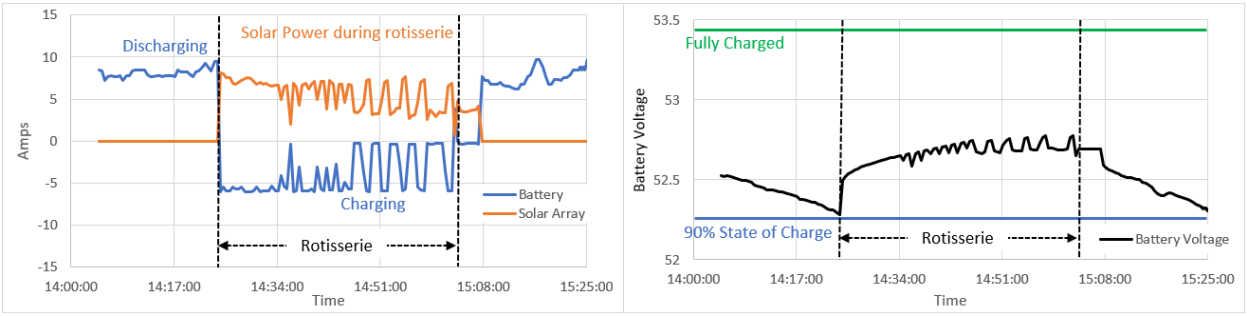


Figure 13. Solar power partially recharged battery during launch rotisserie.

The launch phase completed nominally, with no unexpected telemetry alarms and no unexpected onboard fault protection activity. Figure 13 shows battery voltage and discharge current during the launch. With the solar array in its stowed configuration, the outboard solar array panel had the solar cells facing outward, allowing some power generation to occur. After fairing separation, the second stage entered a rotisserie mode and the spacecraft cycled between power positive (able to lightly recharge the battery) and back to power negative. This rotisserie mode can be seen as small amounts of solar array power and the battery discharge / charge current telemetries going negative, which corresponds to the battery recharging.

Launch vehicle separation attitude and tip-off rates were close to ideal, resulting in the ground having an uninterrupted link to observe continuous telemetry from the start of transmission through entry to the safe mode rotisserie. The two figures below highlight telemetry observed by ground operators in real time during the solar array deployment. The top row of both figures shows progress through the launch behavior, starting with the parallel wait for detumble completion and TWTA warmup. This is followed by the +Y release mechanism firing, the six minute +Y deployment delay, the -Y release mechanism firing, the six minute +Y deployment delay, and completion/transition to safe mode on the right. Figure 14 shows the spacecraft body rates (each axis plotted separate) through the launch activity. The launch vehicle separation (far left) had small rates in all axes, which are driven to near-zero by the launch behavior. The release of both solar array wings is clearly visible in all three axes, as the wings unfurl and latch.

Figure 15 shows the motion of the flip-out side array side panels with the measured angle of the arrays. Both arrays are initially at position zero when stowed (note: telemetry jumps between 0 and 6.28 rad, which is functionally identical but appears as “noisy” data” in plot form). The torque from the deployment of the first flip out panel is sufficient to rotate the whole panel, from 0 to 0.4 radians for the +Y array and 6.28 to 5.88 radians for the -Y array. The movement and latching of the second side panel is in the opposite direction, and rotates the whole wing back from 0.4 radians to 0.12 radians (+Y) and 5.88 to 6.13 radians (-Y). Together with the body rates, this telemetry provided initial confirmation that the array deployment was successful. Body and SADA motion indicated that all release mechanisms were separated and that the side panels were deployed and latched. This was later confirmed with the downlink and analysis of the high rate MIMU data taken during the SA deployment. Finally, the bottom plot shows the battery discharge current. Though the solar arrays would not be actively rotated to the Sun until the post-deployment safe mode entry, the vehicle orientation

and residual rotation rate during the +Y deploy delay provided enough of a solar incidence angle to generate power for the spacecraft to begin charging the batteries.

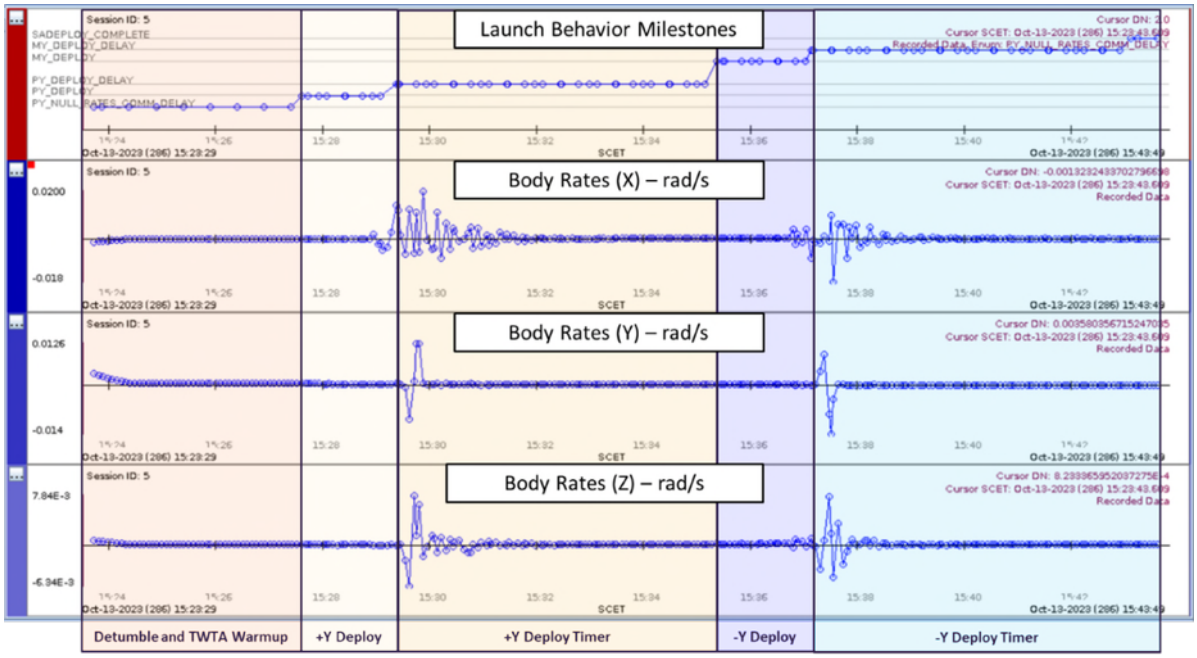


Figure 14. Spacecraft body rates during solar array deployment.

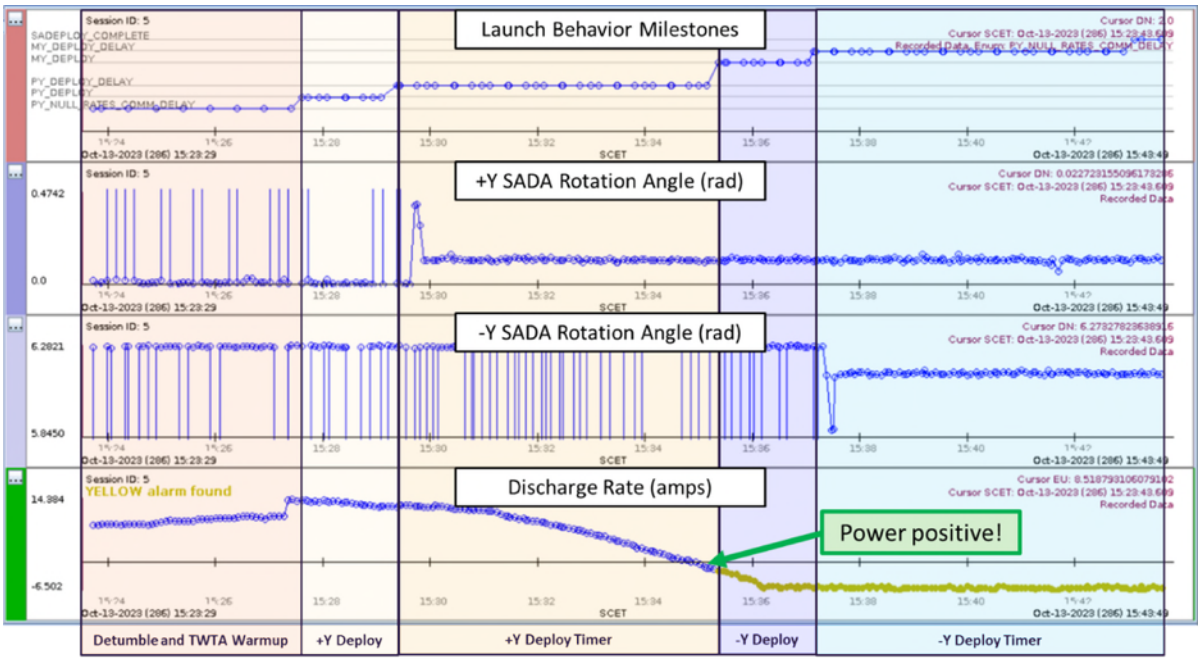


Figure 15. Solar array rotation angle and battery discharge current during solar array deployment.

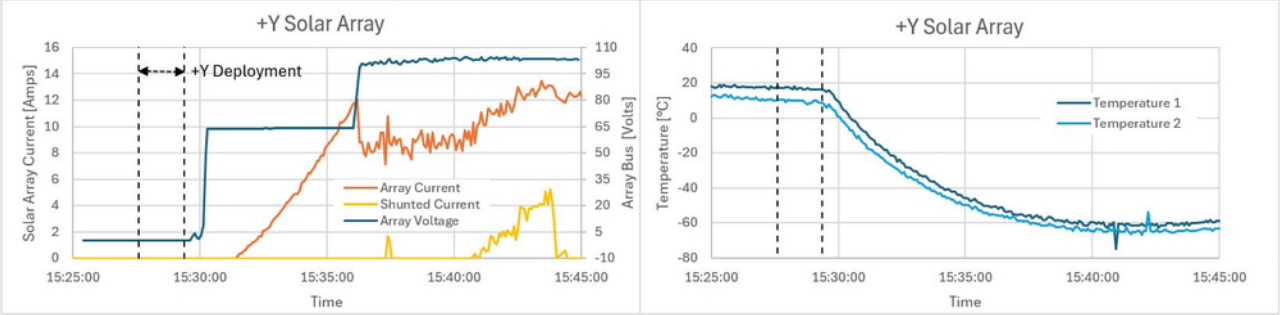


Figure 16 shows the +Y solar array telemetry during this phase. A drop in the array temperature occurred during the time between when the solar array was deployed and the completion of the slew to sun. This increased the solar array voltage to the point where the shunt circuits in the PCU begin regulating the spacecraft power until the arrays begin to warm up again. This demonstrated successful solar array regulation with both the boost converters and the shunt circuits. With deployment complete, the solar arrays returned to their nominal operating temperature, the PCU returned to boost mode, and the spacecraft was power positive, thermally stable, and ready for the remainder of the mission.

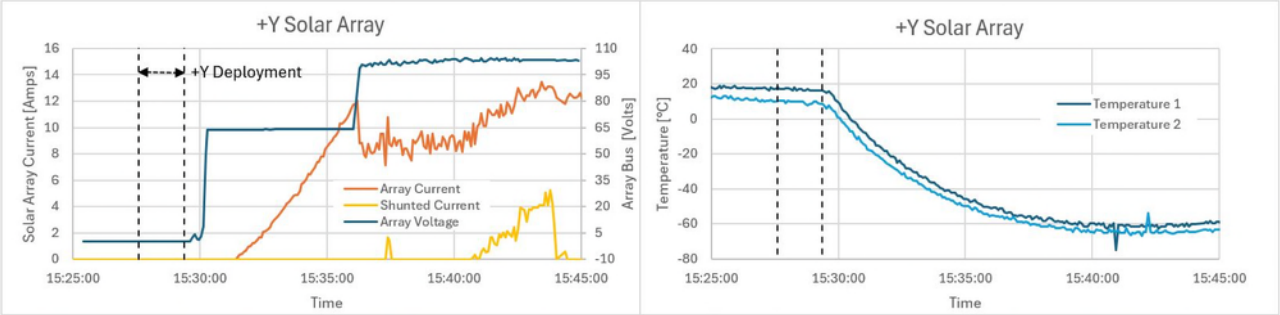


Figure 16. Solar array performance during launch.

B. Cold Gas Propulsion

Psyche's electric propulsion (EP) system is used for primary propulsion in cruise and provides secondary propulsion for momentum management (reaction wheel unloading) in cruise and during some orbital operations around Psyche. However, there are times when the high-power consumption (minimum 1 kW) and delayed start (~5 minutes) of the EP system are problematic for spacecraft operations. Therefore, a nitrogen cold-gas thruster (CGT) propulsion system was installed for time critical maneuvers such as nulling the tipoff rates immediately after launch vehicle separation, for attitude control before the reaction wheels were commissioned, and for safe mode contingencies where the solar arrays may have lost lock with the sun. To reduce disturbances to science operations, use of the cold gas system is also planned for momentum management (wheel unloading) during science orbit operations. Further details on the design of the cold gas system are available in reference 2.

Figure 18 shows a block diagram of the CGT system. There are two independent, fully redundant, thruster strings. Three 82-liter nitrogen tanks are isolated from the rest of the system via latch valves during launch, while series redundant regulators limit thruster inlet pressures to about 360 psi. As shown in Figure 17, thrusters 5/6 and 11/12, operated in pairs, produce rotation about the X axis. The remaining eight thrusters provide rotation about the Y and Z axis. Each thruster is rated at 0.45 N thrust and 70 seconds specific impulse at 360 psi inlet pressure and operates at a maximum 30% duty cycle to avoid overheating. All six thrusters firing on a single string result in a 4.5g/sec flow rate. Analysis shows that over 40 kg of the nitrogen propellant can be used at the rated performance of the system.

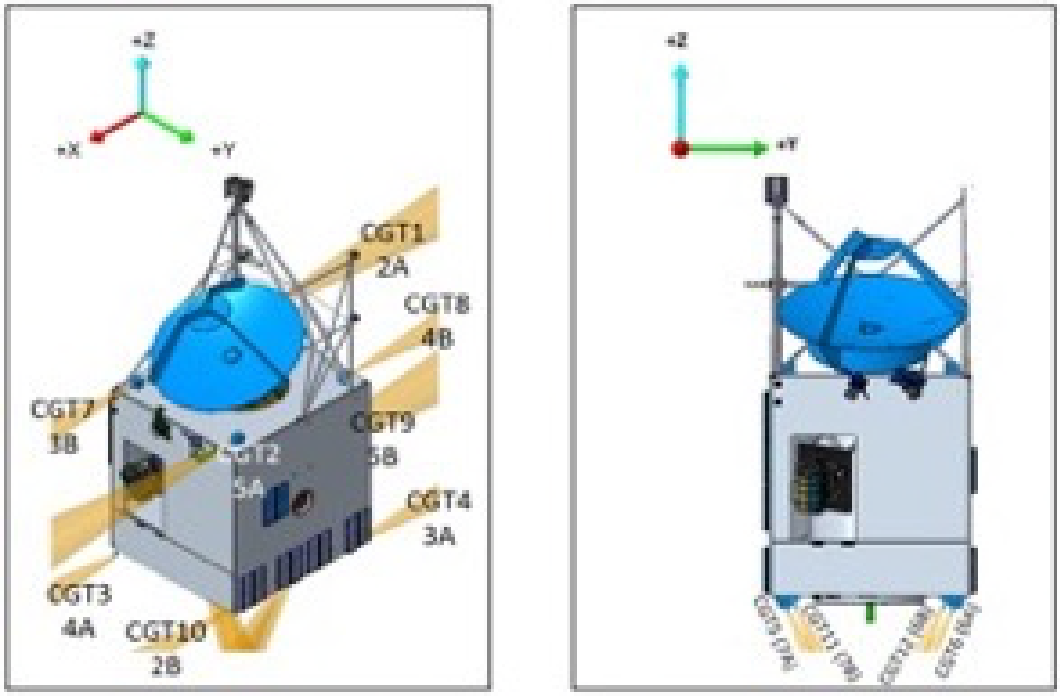


Figure 17. Placement of the cold gas thrusters on the Psyche spacecraft.

The cold gas thrusters were first used immediately after separation to null launch vehicle tipoff rates, which was successful. However, the first use of Side-A CGT system was found to produce a rapid drop in manifold pressure above the thrusters, leading to thruster underperformance. Operationally, the spacecraft autonomously opened the latch valve upstream of the pressure regulator and start firing the thrusters immediately. Figure 19 shows the nitrogen manifold pressure just upstream of the thrusters, and the thruster firing time (by thruster number), versus time. This hard-start produced a sudden large pressure differential across the regulator, which caused a flow restrictor built into the regulator and intended to limit pressure overshoots to close and limit the nitrogen flow rate. Immediately starting thrusting maintained the pressure differential across the largely closed regulator, dropping the pressure and slowing recovery of the regulator. The pressure in the manifold fell to about 230 psi in less than 15 s of operation, reducing the CGT performance by nearly 40%. Fortunately, this firing was still sufficient to null the very low tipoff rates produced during separation and establish 3-axis stabilized control of the spacecraft. Subsequent operation of this Side-A CGT system for attitude control worked perfectly because the pressure across the regulator was now in equilibrium and it operated properly without restricting the flow. Ground testing showed that the regulator requires 15-30 s to recover proper operation after being hit with a large upstream pressure surge from opening the latch valve. A later activity was used to open the latch valve on the Side-B CGT system for several minutes without operating the cold gas thrusters to “prime” the regulator and establish the proper pressures. Both sides of the CGT system are now considered operational and ready for use in the event of a safe mode entry.

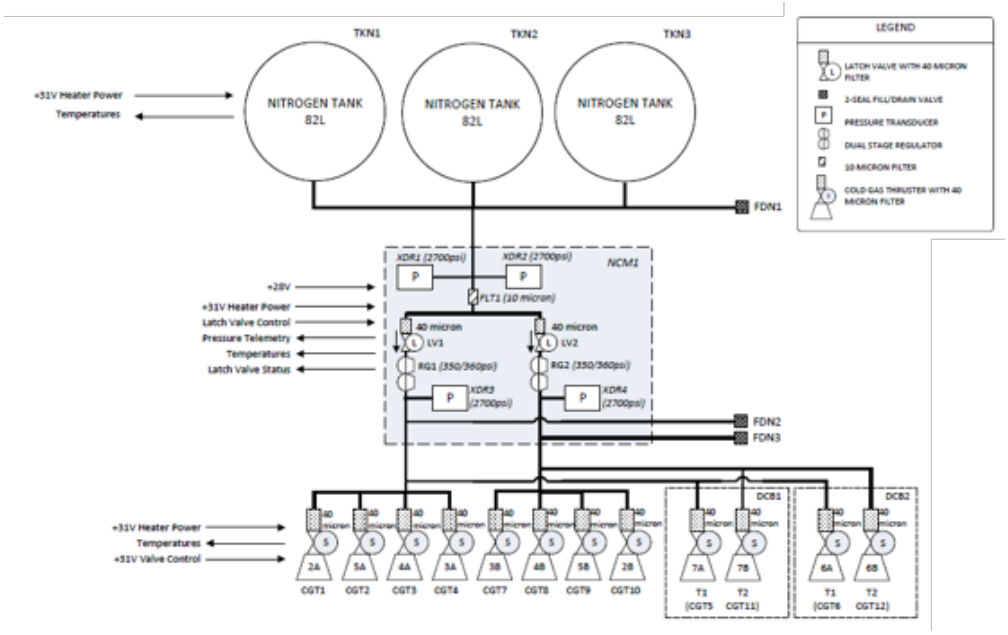


Figure 18. Block diagram of the cold gas propulsion system.

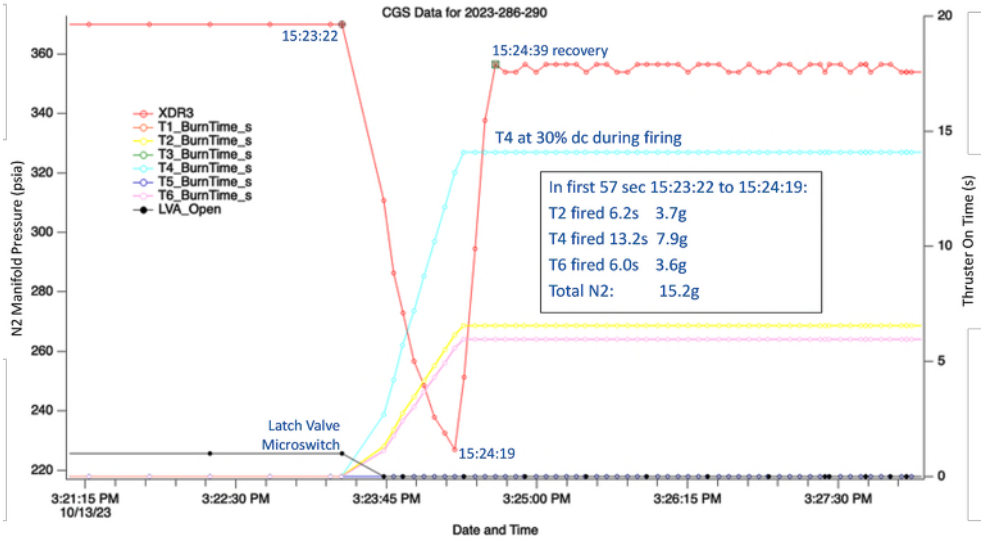


Figure 19. Nitrogen cold gas thruster first use showing the manifold pressure above the thrusters dropping during the thruster firing times.

The cold gas system was used as the only attitude control source on the spacecraft for 2 days until the reaction wheels were enabled, and as the only propulsion system on the spacecraft for 3 weeks until the EP system was commissioned, consuming 0.8 kg. of propellant over that time frame. During that time period the cold gas thrusters were used to mitigate outgassing, solar radiation pressure, spacecraft rotations, and to stabilize the spacecraft during safe mode entries.

V. Spacecraft Initial In-Space Checkout and Commissioning

A. Overview

After completion of launch and deployment activities, the spacecraft entered its initial checkout (ICO) period. The primary objective of ICO was to verify that the spacecraft had survived launch and was ready to autonomously conduct near-continuous cruise thrusting, with a secondary objective to verify proper post-launch operation of as much of the

spacecraft as practical. The time available for ICO was bounded by the need to start cruise thrusting in time to achieve the Mars flyby in May 2026. Therefore, prior to launch, the team developed a fully prioritized ICO plan, to facilitate the descoping or deferral of low priority activities if the plan did not proceed as scheduled.

The prioritization of ICO activities was carried out based on a series of criteria, here posed as a series of questions. 1) Can the test only be executed in space? Or is it confirming a previous ground test result? 2) Is successful activity completion necessary to start Cruise thrusting in a nominal or contingency scenario? 3) Is successful activity completion necessary for DSOC operation? 4) Is the activity a hardware maintenance or parameter/file update activity required to maintain effective operation of the spacecraft? and 5) Is the activity testing only redundant hardware? Based on these criteria, each activity was placed into a category, and then assigned a priority as shown in Table 3.

Test Category	Priority
1) Necessary for Cruise Thrusting	Mandatory
2) DSOC Commissioning	Mandatory
3) Other items related to Cruise Thrusting	In-Plan
4) Necessary Maintenance	In-Plan
5) Science Instrument Testing	In-Plan
6) Other one-time tests/activities	Consider for Plan on Case-by-Case basis
7) Tests of Redundant Hardware	Not in Plan (with exceptions)
8) Other activities	Not in Plan

Table 3. Mapping of ICO test categories to test priorities

The most important objective of ICO was to demonstrate readiness for cruise thrusting, as thrusting is both essential and time critical for the science mission. To this end, the first category encompassed the minimum set of activities necessary to support execution of the Long Duration System Test (LDST), a test which would demonstrate autonomous operation of two electric propulsion Power Processing Unit (PPU) + thruster strings executing a cruise thrusting scenario over a seven day period. Activities in this category were assigned the highest priority and treated as mandatory. Category 2 consisted of the minimum set of activities needed to start operation of the DSOC technology demonstration. DSOC operation was also time critical because of the goal to demonstrate the effectiveness of the system at Earth-Probe distances as low as 0.06 AU. The window for these opportunities started only 18 days after launch. Category 2 activities were given the next highest priority, and treated as mandatory tests.

Category 3 consisted of activities related to thrusting, but not part of Category 1. This included testing of redundant PPU and EP thruster strings as well as contingency activities, such as a window for a second LDST to be conducted if the first LSDT was unsuccessful. It also included accommodation for a pre-LDST flight software (FSW) update, to be conducted only if post-launch testing revealed a major problem that required a software update prior to the start of cruise thrusting. These activities were less time critical than Category 1 and 2 activities, so they were allocated time in the ICO plan at a nominal, and potentially descopeable, priority level. Category 4 included periodic maintenance activities necessary to support the spacecraft (for example, balancing battery cell voltages) and Category 5 consisted of tests of the science instruments. These items are important to the overall mission, but were not generally time critical to execute during the ICO period.

Category 6 focused on remaining tests of primary hardware not covered in the higher categories. Each activity in this category was discussed and prioritized on a case-by-case basis, with an emphasis on collecting data in ICO which had not previously been collected on the ground (for example, measuring the full field of view of the sun sensors). Category 7 consisted of tests of backup hardware. Consistent with its classification as a NASA risk Class B mission[19], Psyche's general policy was to avoid testing redundant hardware in space except under exceptional circumstances. An example of an operational exception was testing of the backup computer, where the need to power on the backup Psyche compute element (PCE) as part of a Category 4 parameter update activity provided a natural opportunity to conduct a full checkout. An example of a technical exception was the testing of the backup SEP chassis avionics, which were part of a multi-unit avionics fault containment region. The backup interfaces to these units were tested in ICO to ensured that they would all be fully operational in the event of an autonomous block swap. In practice, these exceptions resulted in at least partial testing of all elements on the spacecraft except for redundant telecommunications units and cold gas thrusters. Category 8 included all remaining activities, and these generally were not included in the ICO plan.

The ICO test procedures were developed, reviewed, and tested in the testbeds based on the prioritization in the ICO plan. As ICO proceeded, changes were made to both the plan and to the test procedures based on observations of the spacecraft's performance. Formally, the ICO phase lasted through launch +103 days. Figure 20 shows the flow of ICO activities as conducted on the spacecraft.

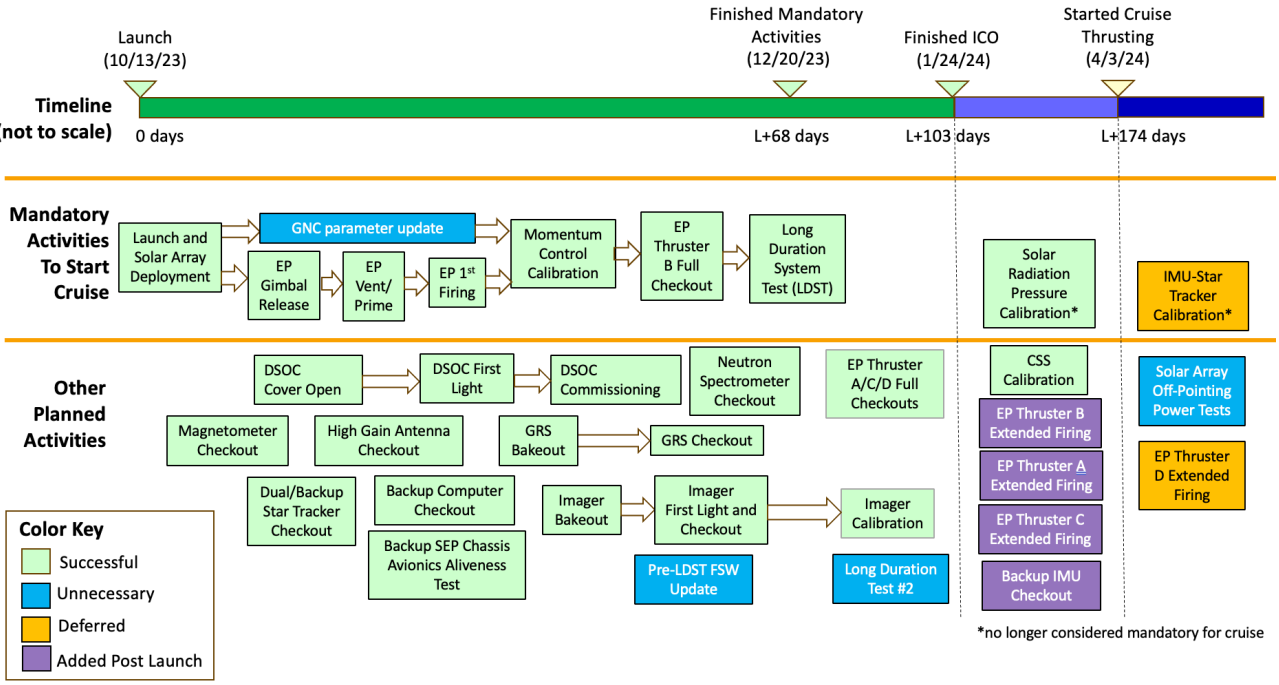


Figure 20. Overview of Psyche's initial checkout activities.
Execution timeline is not to scale

As shown in Figure 20, some commissioning activities were conducted after the end of ICO and before the start of cruise thrusting, and some activities were deferred until later in the mission. Also note that the ICO activities were not necessarily executed in priority order. The rate at which mandatory activities could be executed was limited by staffing constraints and the time needed to fully evaluate each set of test data before moving to the next activity in the chain. To utilize time efficiently, low priority activities were added to the schedule between mandatory activities when DSN passes were available.

The mandatory activities to start cruise thrusting were launch and solar array deployment, priming of the xenon propulsion system, release and deployment of the thruster modules, successful execution of checkout and momentum control maneuvers on the primary EP thrusters, and then execution of the LDST. Two activities designated as mandatory in the pre-launch plan were deferred. An activity to measure the alignment of the MIMU relative to the Star Trackers was deferred based on observed performance of the Guidance and Navigation and Control (GNC) system, and an activity to measure and minimize solar radiation pressure on the spacecraft was deferred based on observations of nominal momentum buildup on the spacecraft. Also based on observed performance, no GNC parameter updates were needed prior to execution of the LDST. The 7-day LDST was successfully executed in December 2023, completing the designated mandatory activities for cruise thrusting. The pre-LDST FSW update and second LDST test were both considered unnecessary after the LDST was successfully completed.

In parallel, DSOC commissioning activities that successfully demonstrated downlink data rates up to 267 Mb/sec between the DSOC system and optical receivers at Palomar Observatory and were completed in December 2023. Lower priority activities, including checkouts of the avionics, payloads, and telecommunications systems, proceeded in parallel, and were successfully completed in January 2024.

As discussed in section II.B, observations of the Hall thrusters during the LDST led to the addition of extended "burn-in" tests for each of the thrusters on the spacecraft. Three of these tests were executed during the interim period between the end of the ICO phase and the start of cruise thrusting, while testing of thruster D was deferred to later in Cruise 1. Observations of solar radiation pressure and sun sensor calibrations were also carried out prior to the start of cruise thrusting. Based on observed performance, MIMU-star tracker calibration activities will not be necessary until the spacecraft approaches Mars in 2026.

Overall, the ICO campaign successfully demonstrated that the Psyche spacecraft was healthy and ready to start cruise thrusting. Cruise thrusting began in April 2024 and will carry on through July 2025. The sections below provide further details on the results of individual ICO test activities by subsystem.

B. Electric Propulsion

As shown in Figure 20, testing of the EP system was at the top of the priority list for ICO. Checkout began twelve days after launch with the release of the DSM launch holddowns and deployment of the DSMs to their nominal flight positions. One week later, the xenon feed system was prepared by evacuating the plenum downstream of the pyrotechnic isolation valves, then opening one of those valves to prime the primary and redundant pressure regulators. The first thruster firing for Psyche, which was also the first operation of Hall thrusters in deep space for any spacecraft, occurred on November 8, 2023. Thruster A ran with PPU 1 at a 3.0 kW discharge power for one hour, then throttled up to the full 4.5 kW power for an additional one hour. Next, Thruster A operated at the momentum management power level of 1.6 kW using PPU 2 for one hour. Following this bakeout, Thruster B was operated with PPU 1 at 1.6 kW and with PPU 2 at 3.0 and 4.5 kW. With this, and with the later bakeouts of Thrusters C and D, all thruster-PPU combinations were successfully operated via all the subsystem electrical harnesses.

SPT-140 thrusters are known to have thrust performance that depends on operating duration and is affected by differences in the space and ground-test environments[20,21]. For this reason, obtaining accurate beginning of life thrust measurements across the whole range of operating powers was a priority for ICO. Each thruster was operated at five power levels between 1.0- and 4.5-kW discharge power under continuous observation by the Deep Space Network with the thrust centerline pointed directly along the Earth line-of-sight. This orientation minimized radiometric thrust measurement uncertainties. A comparison of the results for all thrusters is shown in Figure 21, where the thrust uncertainties are less than ±1 mN 3σ for all measurements. Thrust was linear with discharge power as expected based on ground-test results, and the thrust differences between thrusters was like that seen in the flight thruster ground test measurements. Interestingly, a step decrease in thrust from 277.0 mN to 253.6 mN was observed during Thruster C operation at 4.5 kW. This change is related to thruster "burn-in" phenomena previously seen in Maxar's flight experience and is discussed in much greater detail in Ref. 22. Prior to launch we had expected thrust levels at 4.5 kW to be in the 250-260 mN range after the "burn-in" period was completed, and Thruster C's thrust is consistent with that expectation.

The LDST was successfully completed with the EP system conducting 22 primary mission thrusting operations and 23 momentum management thrusting operations for a total of 140 and 2.5 hours duration, respectively, over the course of the week. There were no major issues seen in any spacecraft systems during EP activities, and the EP system delivered the expected impulse to the spacecraft. However, it was observed on Thruster B that the duration of a thruster burn-in phenomena sometimes referred to as "mode hopping" was longer than previously seen in Maxar's flight experience. This did not impact the overall success of the LDST, but led to the performance of additional thruster activities after ICO. These activities included operation of thrusters at different power levels and magnet currents, as well as extended operation at the full-power 4.5 kW operating point, all of which are described in detail in Ref. 22.

By the end of the ICO period, all EP subsystem hardware had been operated and proven to be functional across the full operating power range with one significant caveat: Thruster B had shown out-of-family burn-in behavior. This behavior was addressed in extended operational tests conducted before the start of cruise thrusting. A summary of thruster operations performed during ICO is shown in Table 4. PPU operation was split between 62 cycles and 340 hours for PPU 1 and 71 cycles and 22 hours for PPU 2. These checkout activities consumed about 1.7% of the total xenon load of 1085.5 kg initially loaded on the spacecraft.

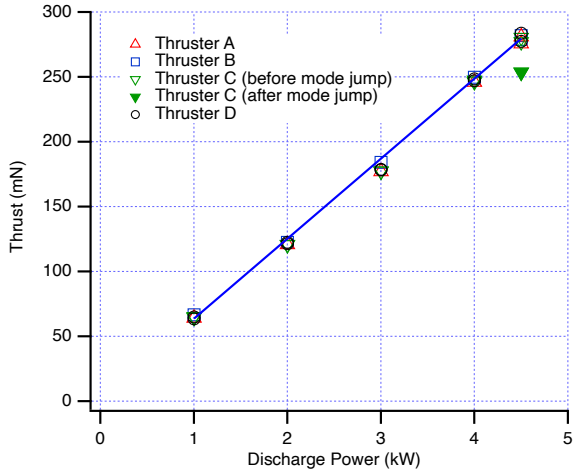


Figure 21. Comparison of electric thruster calibration results.

	Number of Starts	Total Operation Duration (hours)	Propellant Consumed (kg)
Thruster A	46	106	5.1
Thruster B	49	170	9.1
Thruster C	17	73	4.1
Thruster D	21	8	0.3
Total	**133**	**357**	**18.5**

Table 4. Summary of electric thruster operations in ICO

C. Power

The performance of the spacecraft's solar power system is critical for deep-space missions using electric propulsion as primary propulsion. Because this was the first flight of Psyche's unique power system architecture, verifying that it was operating properly was a key priority for ICO. Psyche's power subsystem is derived from Maxar's heritage power system for GEO spacecraft, with two key differences: changes to the electrical layout of the solar array, and the addition of boost converters in the PCU. The resulting design aims to reduce power train losses to the lowest possible level when the spacecraft is at its furthest distance from the sun. A block diagram of the power subsystem is shown in Figure 22.

As shown in Figure 2, Psyche has two solar array wings, each of which has five panels. As the spacecraft travels away from Earth, the voltage generated by the solar cells varies strongly with changes in temperature and light intensity. Psyche's solar cells are wired in series strings that provide power at 60 to 105 volts, depending on distance from Earth. The solar arrays were designed to provide 21 kW of power at Earth and 2.6kW of power at Psyche when at 3.11 AU solar distance. By comparison, maximum spacecraft load while thrusting is expected to be approximately 5.5kW. To aid in performance measuring and trending, the solar array provides two temperature sensors and three solar cells on each wing dedicated to measuring Open Circuit Voltage (Voc) and Short Circuit Current (Isc). The Isc sensors approximate short-circuit current by shorting a single solar cell across a 3.3-ohm resistor or 20-ohm resistor.

Psyche's PCU incorporates converters that boost voltage from the solar array to create a regulated 100V bus. Because Psyche's lowest expected solar array voltage is enveloped by the input voltage range of Maxar's GEO-heritage battery discharge converters, Psyche's PCU repurposes these discharge converters into boost converters without modification. The result, a combined "boost-regulated" and "shunt-regulated" architecture, is shown in Figure 23.

Maxar's GEO spacecraft regulate voltage with a shunt limiter, which works most efficiently when the peak power voltage of the array is close to the voltage of the regulated bus. Due to the wide variation in array voltage, a conventional shunt regulated power system would have been extremely inefficient for Psyche. Instead, at beginning of life, when solar fluence and array output power are highest, discharge converters boost the array output voltage to a regulated bus voltage of 100V. The resulting power conversion losses can be tolerated easily by the spacecraft, as many kilowatts of excess power are available. As the spacecraft moves away from the sun, the peak power voltage rises as the arrays cool. The array is designed so its peak power producing voltage, including solar array degradation and low intensity illumination effects, will reach 100 V at end of mission. At this point, because the array voltage matches the bus voltage, the power conversion boost stage is replaced with a diode pass-through circuit from the solar array to the PPU. This effectively increases the boost converter efficiency from 97% to 99% as the spacecraft approaches Psyche, maximizing power available for science operations. The sequential shunt unit provides bus regulation and overvoltage protection when the solar array voltage exceeds 103V.

This SEP power train design trades higher losses when excess power is abundantly available for higher efficiency when the available solar power is low, while maximizing the use of existing, flight proven hardware from Maxar's commercial GEO spacecraft design to create a system that works on mission trajectories of varying solar intensity.

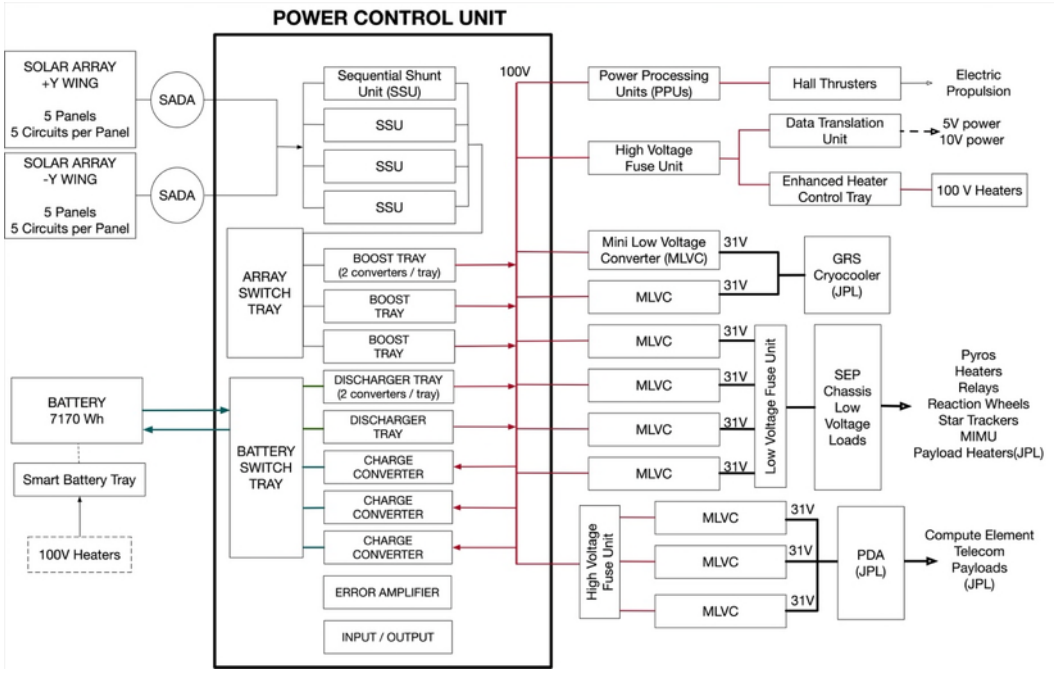


Figure 22: Psyche power subsystem block diagram.

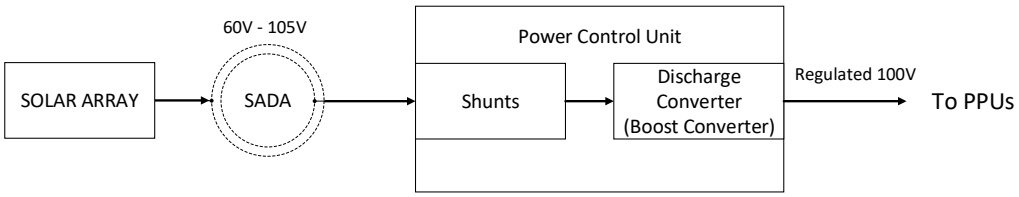


Figure 23. Discharge converters from MAXAR's GEO heritage power control unit boost solar array voltage to create 100 V regulated power for the PPU.

During eclipses, a single lithium-ion battery provides power to the spacecraft. Psyche's battery is a Maxar heritage battery with a single string of cells. Each cell incorporates a cell shorting device which can bypass a failed cell. The battery voltage is up converted to a regulated 100V bus by discharge converters in the PCU, which are separate from those used to boost solar array voltage. To charge the battery during sunlight operations, a set of charge converters in the PCU down-convert regulated 100V bus power to provide charge power. Charge control is provided by autonomous battery management software in the PCE.

Because the solar array produced significantly more power than the spacecraft consumed during ICO, solar array peak power production was difficult to measure directly. However, we did monitor solar array temperature and use the Voc and Isc sensors to compare actual vs. pre-launch performance predictions. As shown in Figure 24, for a given solar distance, the solar array temperature has been trending approximately 10C colder than expected in the pre-launch solar array power model. Open Circuit Voltage is also trending high, which is consistent with the colder measured temperatures. Short circuit current, however, is matching expectations. Although we cannot yet directly measure peak power, if we assume the predicted fill factor has not changed, then colder temperatures imply higher available power as a function of solar distance.

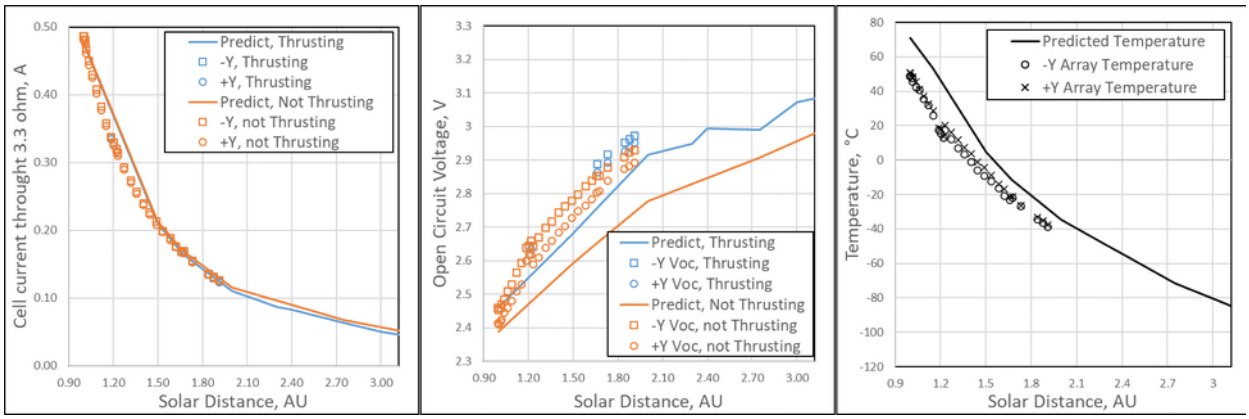


Figure 24. Solar cell temperature and open circuit voltage (Voc) are trending colder than expected, while short circuit current (Isc) closely matches expectation.

A result of this trend is that the power system will begin shunting power earlier in the mission than originally planned. As of May 2024, because the array voltage is higher than originally predicted, the power system has transitioned out of boost mode and is generally operating in bypass mode (above the voltage where boost converters are active but below the level where shunt regulation takes over), even while the thrusters are operational. Based on current trends, it is anticipated that the power system will regularly enter shunt regulation mode when the spacecraft passes 2.4 AU Sun-Probe distance during the first cruise leg, before the Mars Gravity Assist. This is much earlier than originally planned, but for now, predictions of power generated by the array show no impact to the overall performance of the spacecraft and its ability to operate its electric propulsion systems is consistent with the baseline cruise trajectory. The team will continue to trend array performance and monitor the impact to the mission as the array temperature continues to change over time.

Spacecraft loads were measured during ICO and early Cruise, and compared to predicted values. Many spacecraft loads, such as heaters and reaction wheels, cycle on and off. Pre-launch, we had only a limited opportunity to measure typical thermal loads on the spacecraft during thermal vacuum system test. But because there are many patterns, and those patterns may or may not be in phase with one another, the total system load appears to vary randomly above and below the predicted average. Psyche's operations team measured the average system load during various activities by collecting 2-3 hours of total load current and bus voltage telemetry, multiplied voltage by current at each timestep, and then calculated an average power over the entire epoch. Each 2-3 hour epoch was selected to measure a period of time where the spacecraft configuration was static, and consistent with typical operations. Psyche trends average spacecraft load for both non-thrusting and thrusting cases. Figure 25 shows average loads while thrusting with various thruster output power levels. The spacecraft load shown below is inclusive of flight system, science payloads, and electric propulsion.

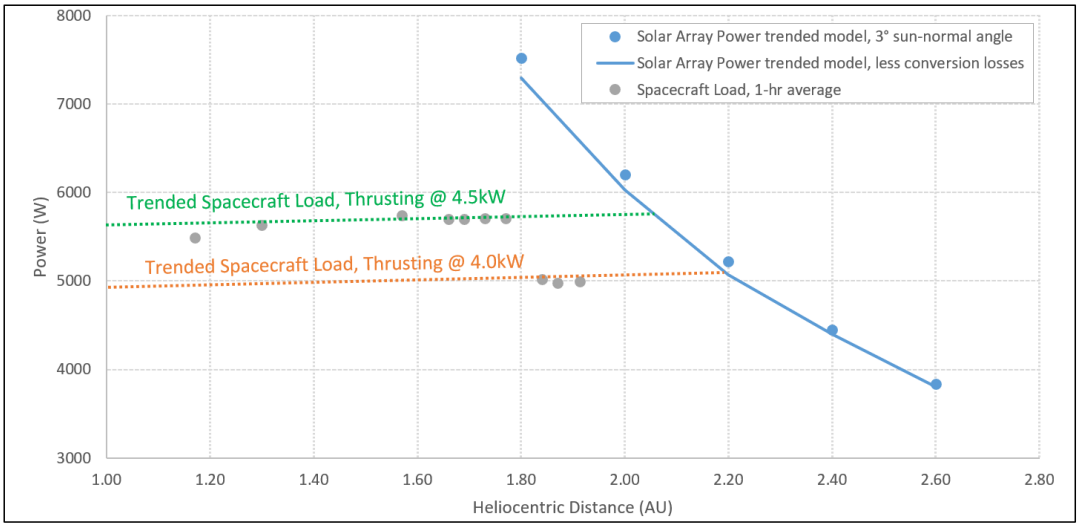


Figure 25. Trended spacecraft power loads vs. predicted solar power production.

D. Guidance, Navigation and Control - Momentum Management

Psyche is the first mission to conduct extended firings of Hall thrusters in a fixed-inertial direction on an ongoing basis. Momentum management is a key challenge, and as shown in Figure 20, demonstrating Psyche's ability to successfully manage momentum using a combination of RWA's and EP maneuvers during LDST was mandatory for the success of ICO.

Hall effect thrusters produce torques via two distinct mechanisms: 1) "Transverse" torques orthogonal to the thrust axis based on the alignment of the net thrust vector and the spacecraft center-of-mass and 2) "Swirl" torque around the thrust axis resulting from the interaction of the thruster's magnetic field with beam ions. As with many EP spacecraft, Psyche's thrusters are mounted on two-axis gimbals which compensate for the transverse torques. During cruise thrusting, Psyche faces unique challenges managing swirl torque relative to other EP missions because:

- Psyche's mission profile in the cruise phase requires thrusting in a nearly-inertial direction over extended periods of time. Spacecraft orbiting the Earth follow time-varying attitude and thrust profiles and can compensate for and/or unload swirl momentum when it rotates into a transverse axis in the spacecraft's body frame.

- Psyche's Hall-effect thrusters produce an order of magnitude more swirl torque than gridded ion thrusters used on previous deep space missions like Dawn and Bepi-Colombo. As a result, compensating for the torque using only cold gas would require prohibitively high quantities of propellant.

- Psyche can operate only one thruster at a time. Spacecraft which can simultaneously operate multiple thrusters, as is done during electric orbit raising on Maxar's 1300 series spacecraft, can use the transverse torques from one thruster to compensate for the swirl torque of another thruster.

The strategy Psyche has adopted to manage momentum is to continually close the loop on the two transverse axes using the EP gimbals during thrusting while allowing swirl torque momentum to accumulate in the RWA's. Once the stored momentum reaches the system's capacity limit, the spacecraft stops thrusting, turns, and executes a short-duration low-power EP momentum maneuver to compensate for the swirl torque. This strategy achieves long-term, closed loop momentum control of all three axes. To minimize time associated with turns and motion of the EP gimbals, a thruster on the opposite side of the spacecraft is used for the momentum management maneuver. Additionally, rather than "unload" the momentum to zero, momentum is "preloaded" opposite the swirl torque direction in the next thrust arc each time a momentum maneuver is performed. Figure 26 shows the preload maneuver in a configuration where Thruster B is used for mission thrusting and Thruster A is used for momentum maneuvers. This matches the configuration used in the LDST. Figure 27 is a timeline view of this sequence showing a full momentum management cycle.

Executing mission thrusting with turn-and-burn EP momentum management involves a mixture of ground commands, onboard sequence logic, and flight software capabilities. Ground operators determine the start time and direction of primary mission thrust maneuvers and swirl preload maneuvers weeks in advance of execution and then implement these choices into onboard sequences. Onboard sequence logic determines the magnitude and direction of EP momentum maneuvers and their termination times during execution. Flight software maintains attitude control of the spacecraft and does closed loop management of transverse momentum using the EP gimbals during mission thrusting. The spacecraft remains fully under three-axis RWA attitude control at all times, including during all EP maneuvers.

Although in principle this process could be automated to occur autonomously under spacecraft control on an as-needed basis, various constraints placed on the spacecraft operations schedule mean that the timing of these maneuvers may not be freely selected onboard. For example, every week during thrust periods the spacecraft communicates with the DSN for several hours on a low gain antenna for thrust verification. It is not acceptable for a momentum maneuver to occur during this time because turning the spacecraft could point the selected antenna away from Earth. To accommodate these constraints, spacecraft operators are tasked with planning the cadence of preloads to ensure that the momentum stored in the RWA's does not exceed limits while also deconflicting with various other activities.

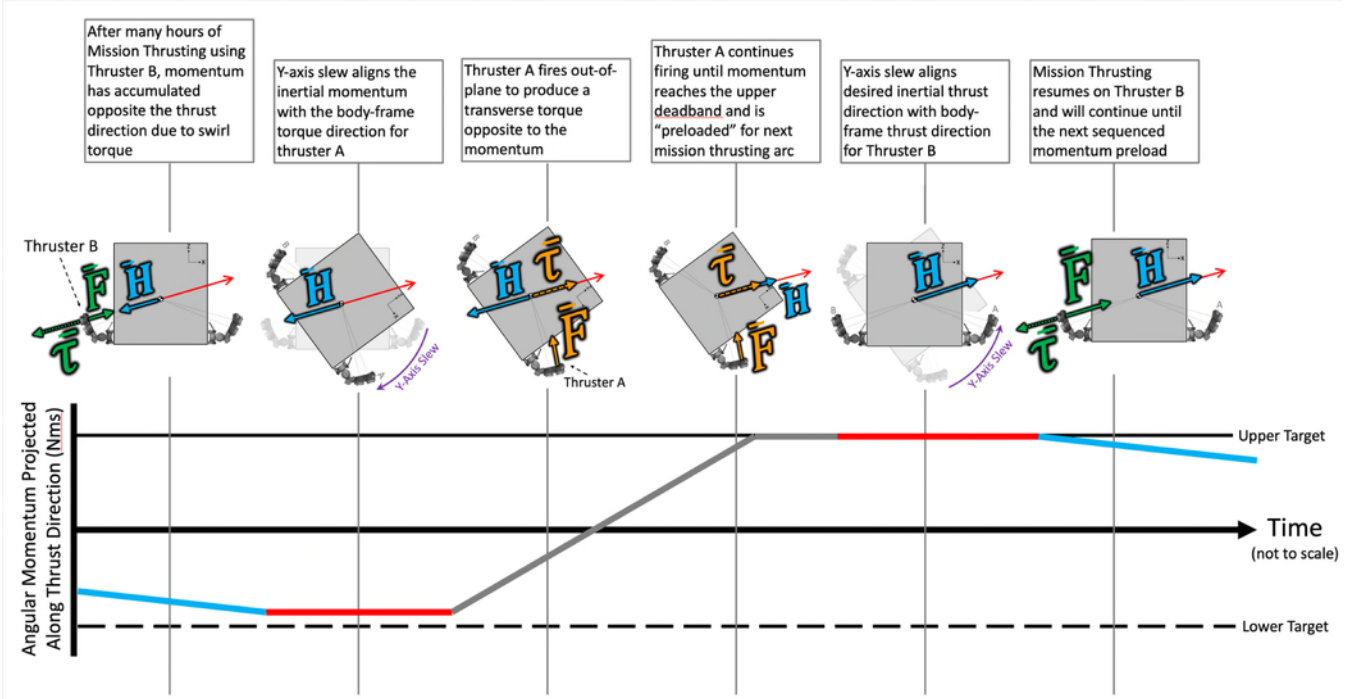


Figure 26. Illustration of turn-and-burn EP momentum management. Example thruster selection matches LDST configuration: Thruster B for mission thrusting and Thruster A for momentum maneuvers.
H = System Momentum, F = Thrust Force, τ = Torque, Red Arrow = Mission Thrusting Direction

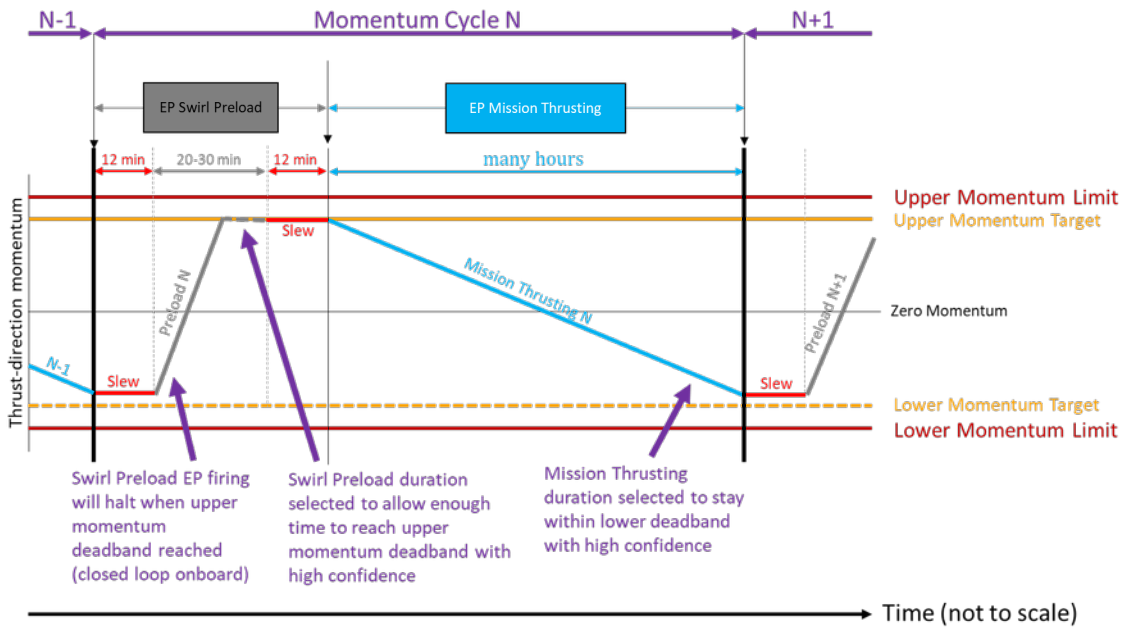


Figure 27. Timeline view of a single momentum cycle (one preload plus one mission thrusting arc).

During the cruise phase, Psyche uses month-long sequences of absolute timed commands, which means that this planning must occur about two months in advance of execution. To facilitate this process, thrusting is planned in self-contained, modular units known as "Mission Thrust Periods" or MTPs. Each MTP begins and ends with zero net momentum on the spacecraft, all thrusting is in nearly the same inertial direction, and preloads within each MTP follow the same cadence. The cadence of momentum maneuvers is calculated based on the predicted thrust level, predicted swirl

torque, spacecraft physical properties, and other factors during each MTP. Finally, a set of rules is used by scheduling software to place specific momentum maneuvers relative to other spacecraft activities. Figure 28 shows some salient features of an MTP.

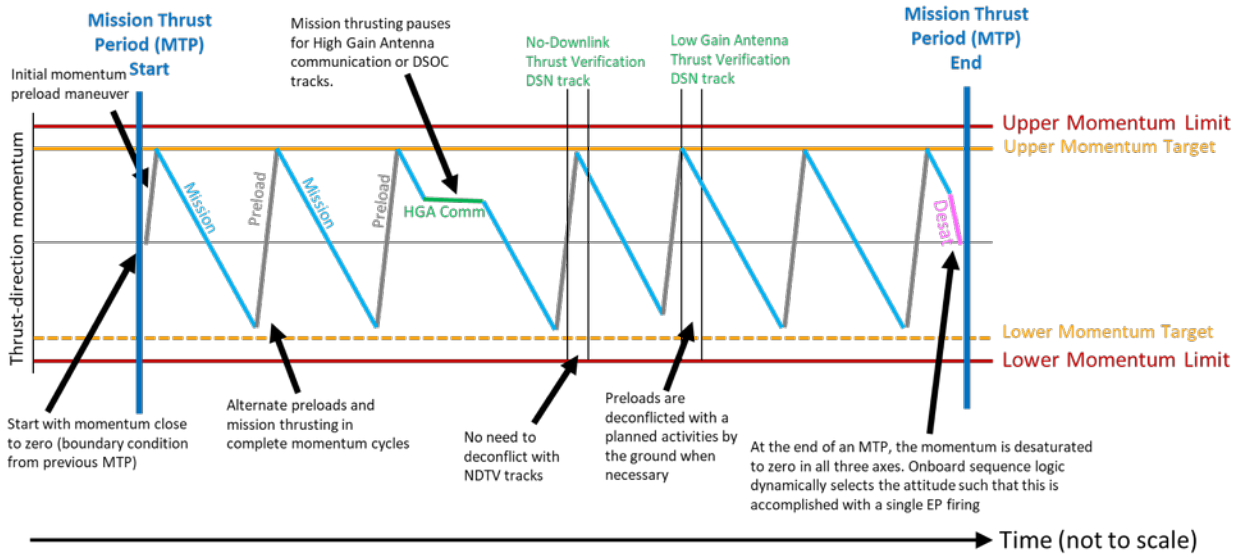


Figure 28. Illustration of Mission Thrust Period.

Figure 29 shows the actual thrust-direction momentum for the week-long LDST, which was planned as a single MTP and tested all the momentum management activities needed for cruise thrusting. Successful execution of LDST validated the overall momentum management strategy as well as the spacecraft and the operation team's ability to effectively execute extended cruise thrusting. However, the results from this test and other ICO activities have resulted in adjustments to this strategy which affect the mission's cruise thrust duty cycle going forward. These results are summarized below.

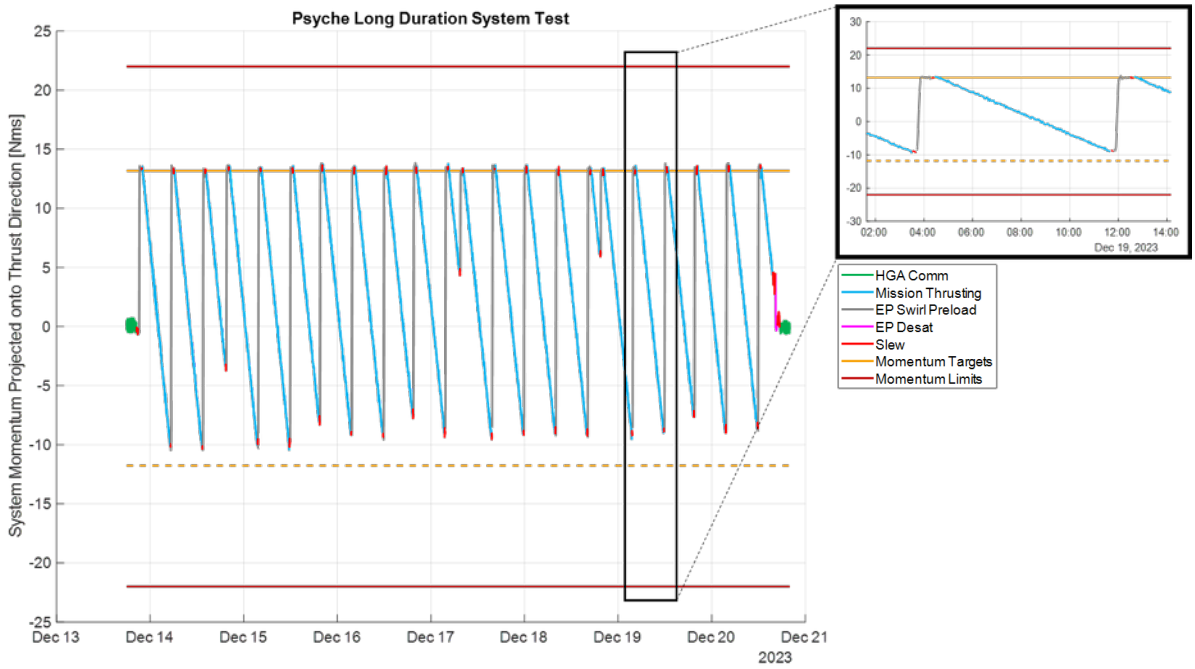


Figure 29. Momentum time-history for LDST with inset zoom on one momentum cycle.

A first-order driver on the time between preloads is the predicted swirl torque. Accurately measuring the swirl torque from a Hall thruster is extremely difficult to do on the ground. This value has never been measured in ground testing, was never accurately measured by any missions flying SPT-140's in Earth orbit, and proved difficult to estimate

analytically pre-launch. Early in the conceptual design phase of the mission, the swirl torque was estimated with a simple first-principles method to be 0.267 mN-m at 4.5 kW discharge power.[23] This estimate was later raised to 0.66 mN-m based on a combination of computational modeling and ground measurements of ion speed distributions[2]. Calibration activities conducted during ICO measured an estimated swirl torque value of up to 0.92 mN-m across the four thrusters at maximum power. This is about 40% higher than expected pre-launch and about 250% higher than the original estimate from the concept phase.

An additional complication arose when higher-than-expected spacecraft attitude oscillations were observed during ICO, when the reaction wheel speeds were near zero shortly after launch. Because this was a potential lifetime and performance concern for the RWA's, with an abundance of caution, the project chose to avoid operating the reaction wheels at low speeds while the issue was investigated further. This resulted in the system momentum limit being reduced to ±22 Nms from an expected pre-launch value of about ±35 Nms.

The combination of these two issues, the practical need to add momentum margin to account for operational uncertainty, plus other conservative choices made prior to completing checkouts and data analysis increased the cadence of preloads required such that they occurred every ~8 hours during the LDST rather than every ~3 days as anticipated in during Psyche's conceptual design phase. Maintaining this cadence going forward would significantly and negatively impact the time available for thrusting by lowering the mission's effective cruise thrust duty cycle. To avoid this outcome, and looking ahead to the rest of the cruise phase, the operations team is following multiple avenues of improvement which are expected to roughly double the time between preloads. Notably, the restriction on avoiding the low speed regime of the reaction wheels will be removed in combination with an increase in the reaction wheel heater setpoints to improve lubricant life. This will substantially increase the system momentum storage available on the spacecraft.

The first few months of Psyche operations have shown that the turn-and-burn strategy for managing momentum is effective, but operationally complex, requiring a substantial fraction of the effort spent during each month's planning time. Future missions should carefully consider whether hardware mitigations (such as always firing two thrusters, or using a single thruster with a reversible magnetic field) or changes to the GNC flight software algorithm (such as the use of onboard closed-loop "spiral" or "zigzag" thrusting patterns) can be used to reduce the operational complexity associated with swirl torque management when using Hall thrusters for interplanetary missions.

E. Thermal

Thermal control on the Psyche spacecraft is achieved using both passive and active elements. Passive elements include thermal blankets, coatings, shields, radiators, louvers, and heat pipes. Active elements consist of autonomously controlled heaters. Thermistors provide temperature data that are used for thermal control and health monitoring.

Psyche's thermal dissipation loads are relatively modest, dominated by bus units, and are easily accommodated on a simplified version of Maxar's smallest GEO spacecraft radiator panels through the removal of heat via thermal control hardware and Optical Solar Reflectors (OSRs). Beyond the dramatically reduced thermal control area, Psyche's thermal design also uses thermal louvers to keep the spacecraft from becoming too cold when far away from the Sun. Thermal louvers are passive mechanical shutters placed over OSRs that open to provide a clear view to space when warm, and close to limit radiative cooling when cold.[24]

The thermal design of the spacecraft has five thermally separated zones: the main body, solar arrays, antennas, batteries, and tower. These zones are isolated from each other as much as possible to reduce heat transfer between them, allowing for thermal control to be performed independently for each zone. In the spacecraft main body, thermal control is largely passive and augmented with heat pipes in the equipment panels. All the major equipment panels and thermal zones, except for the battery radiator, are shown in Figure 30. On the equipment panels, there are additional zones isolating main bus equipment from the MIMUs and from the DSOC payload radiator area. Battery thermal control is also achieved passively with jumper heat pipes to the dedicated battery radiator.

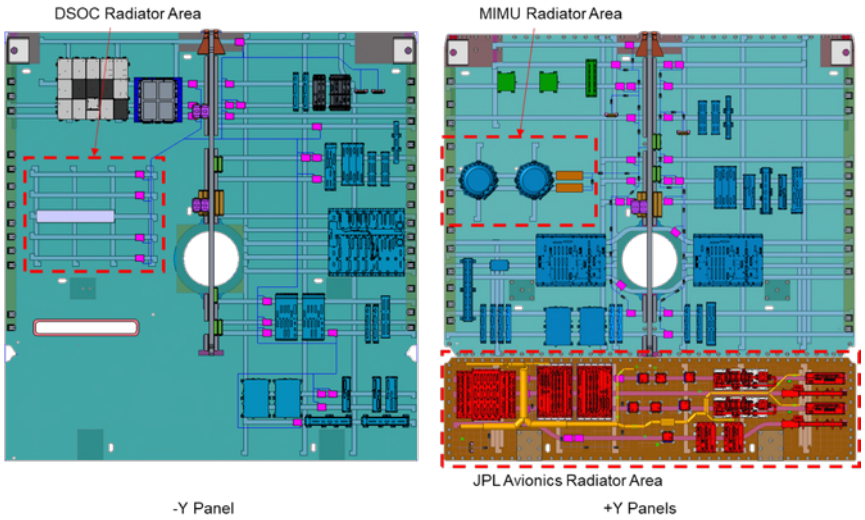


Figure 30. Main body equipment panels showing special radiator areas for DSOC, MIMUs, JPL equipment. ***Other areas support a matrix of heat pipes for the SEP chassis avionics.***

The Psyche spacecraft experiences its hottest conditions at the start of the mission when it is at 1 AU solar distance under cold gas thrusting conditions. The radiator area has been sized to keep the panels (and therefore the highly dissipative units) under their hot temperature limits during this early phase of the mission. The radiator surfaces are covered by the louvers, which close as temperatures decrease on the way out to Psyche. This allows the spacecraft to minimize the heater power required to keep the equipment above the lower component temperature limits. This approach allows the heater power to be kept under 150W while under nominal operations at Psyche. Initial launch conditions show that the radiators maintain temperature under low power usage, and increase in temperature as high dissipation units operate during the checkout process as seen in Figure 31 and Figure 32.

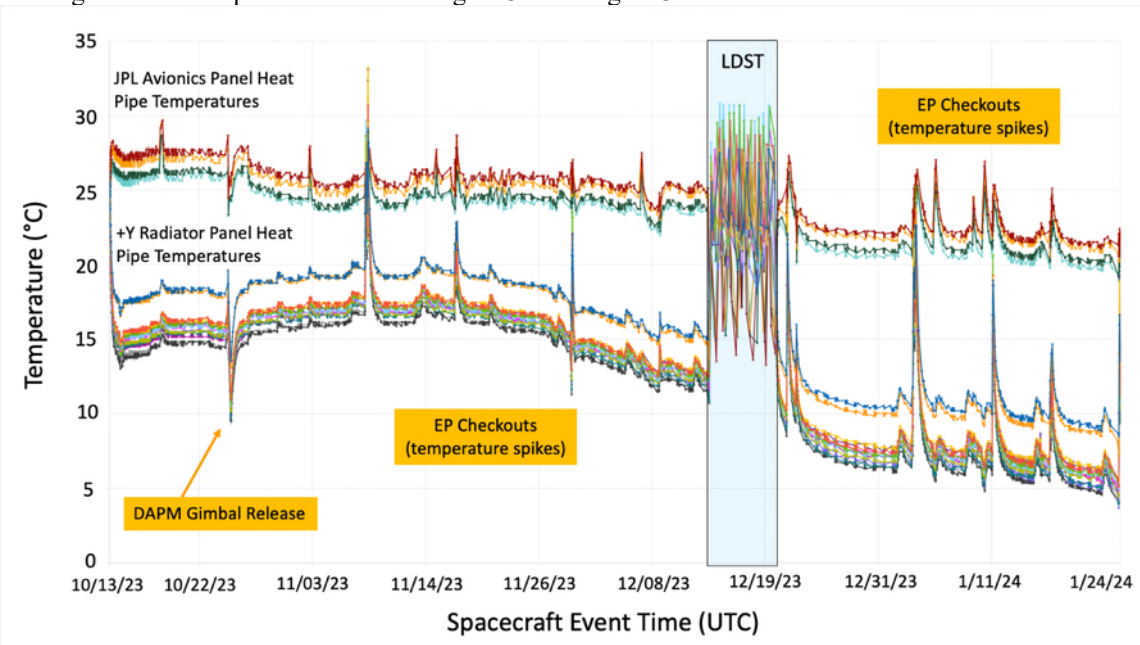


Figure 31. +Y and JPL avionics panel heat pipe temperatures during ICO.

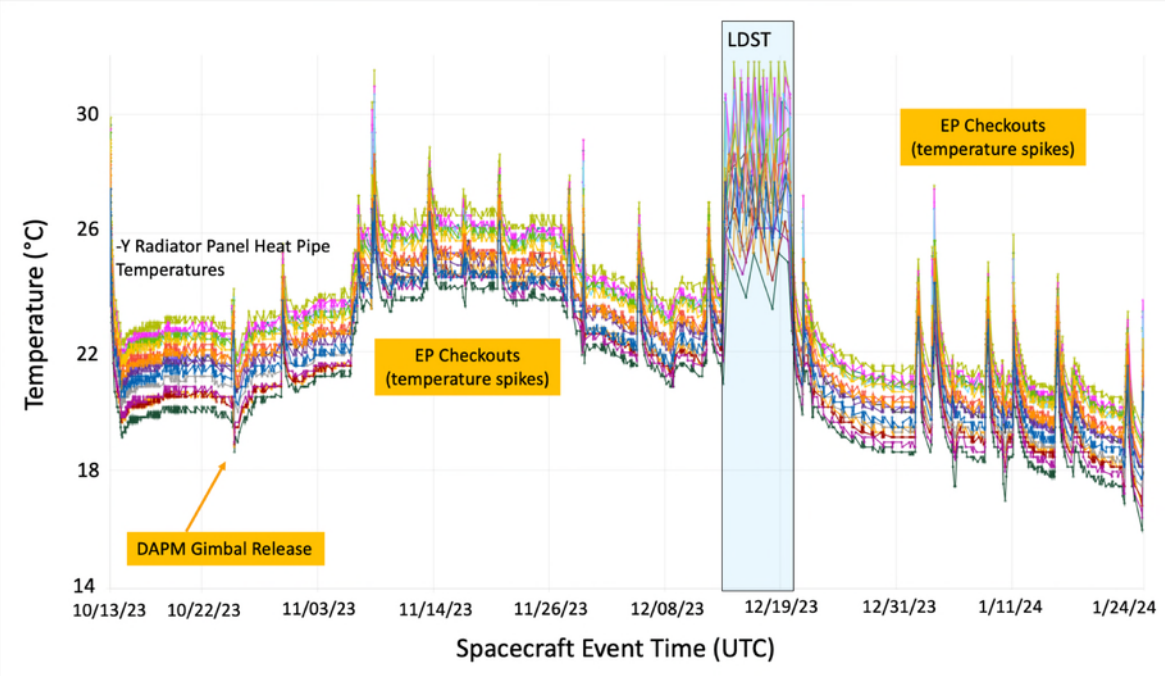


Figure 32. -Y panel heat pipe temperatures during ICO.

Optical Solar Reflectors (OSRs) are mounted on the +Y and -Y equipment panels, JPL radiator panels, battery radiator, and Star Tracker radiator panels. OSRs minimize the solar energy absorbed and maximize the emission of infrared energy to space, thereby cooling the radiator panels. With these faces used as radiators, solar heating effects on main body temperature variations are minimized. Operational restrictions also keep sun off the Y panels during cruise to maximize radiative capability while minimizing heater power required at Psyche. As seen in Figure 33, during much of ICO, the sun was on the +X panel, so the OSRs were critical to maintaining the Star Trackers at acceptable temperatures under constant solar loads. During events with the sun removed from the Star Trackers, they cooled back down as the OSR radiators were exposed to deep space.

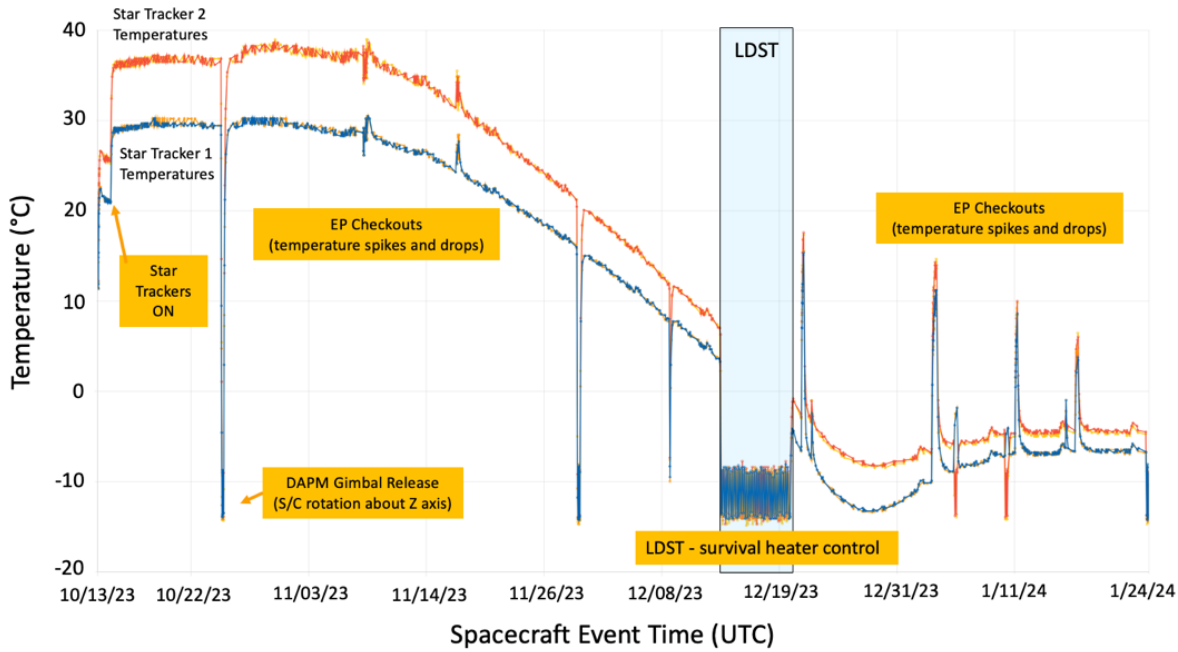


Figure 33. Star tracker temperature measurements during ICO.

Heat pipes are embedded in the +Y and -Y equipment panels, JPL panel, and battery panel to minimize temperature variations and to transport heat from high-dissipation units to cooler areas of the panel and external OSR radiator surfaces. Heat pipes consist of ammonia filled tubes that distribute heat and thus provide additional paths for heat transfer to take full advantage of the radiator panels. There are jumper heat pipes that go from the DSOC payload to the -Y panel to reject DSOC dissipation during the first year of cruise, and jumper heat pipes from the battery to the battery radiator as mentioned previously. This can be seen in Figure 34 during the operation of DSOC as the units are energized and dissipation is routed from DSOC to dedicated radiators on the -Y panel.

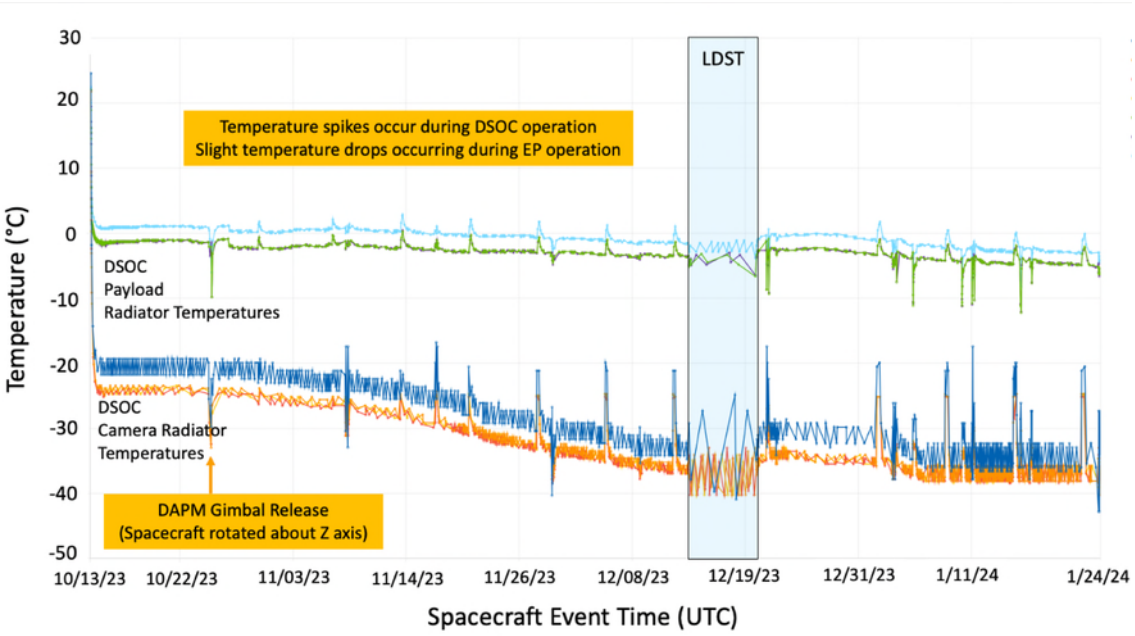


Figure 34. DSOC related temperatures during ICO.

During the Long Duration System Test, one thruster (Thruster B with PPU1) was fired for a period of a week. The maximum thruster and PPU temperatures are shown in Figure 35. The PPUs, mounted to the heat pipe panel, distributed heat and maintained their baseplate temperature to a maximum of 35 °C. The thermal design of Thruster B had a maximum firing temperature of 175 °C. The temperatures of PPU 2 and Thruster A were lower because they were non-operational for this test. These results are in line with what is expected for firing close to Earth, and temperatures will decline as the spacecraft moves farther out into the solar system.

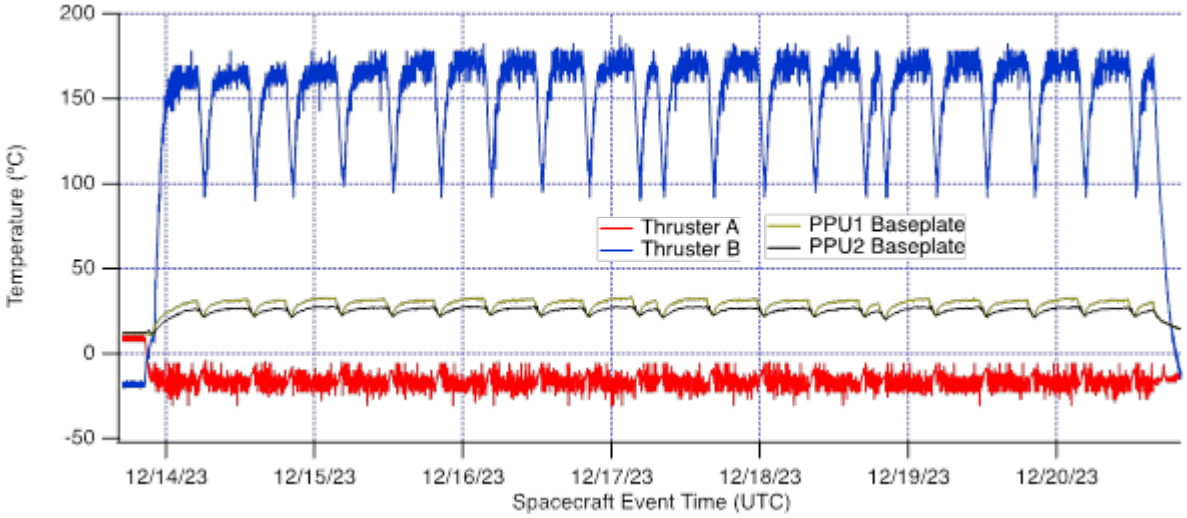


Figure 35. Thruster and PPU temperatures observed during the Long Duration System Test.

Through ICO, the overall thermal subsystem has demonstrated excellent performance. The Psyche thermal subsystem is performing to specification, and it is expected to continue performing well for the remainder of the mission.

F. Science Instruments

Although operation of Psyche's science instruments was not required to ensure readiness for the cruise thrusting, science instrument checkouts were included in the ICO plans with several important goals. Besides establishing proof that the instrumentation survived launch, some critical calibrations and opportunities could only be uniquely done early in the mission. It was also beneficial to establish baseline measurements in space, early in the spacecraft's lifetime, to compare against future performance.

It was important to power on the Magnetometer as early as possible, before other subsystem test activities, to allow the collection of data that would characterize the spacecraft itself. Each unit's individual contribution to the spacecraft magnetic field was measured as it was turned on. Information about component magnetic fields is compiled in a look-up table that can be used to extract the spacecraft field from science measurements. Magnetometer data can also be a useful diagnostic tool for identifying electromagnetic (EM) signatures for diagnostics of spacecraft behavior and anomaly resolution. For this reason, the Magnetometer was powered on October 17, five days after the spacecraft launched, and remains on today autonomously delivering continuous magnetic field data.

The top level requirement to determine the asteroid's magnetic moment has been decomposed into requirements on magnetometer instrument uncertainty, and knowledge of the spacecraft's magnetic field. Based on test results, the team confirmed that the instrument sensitivity is within requirements, with a root mean square field per axis of ~25 pT. In addition, the spacecraft field has been confirmed to be very low, with the DC magnetic field of ~20 nT at the inboard sensor and no detectable AC fields. This result confirms that the pre-flight EMI/EMC magnetic cleanliness efforts conducted during spacecraft integration were successfully able to minimize residual magnetic fields on the spacecraft.

Checkout activities included collecting magnetic field data at both 50 Hz and 10 Hz sampling frequencies and assessing data while in three different instrument excitation frequency settings to assess interference from the reaction wheel assemblies (RWAs). Characterization at all settings indicated no interference or data degradation. Figure 36 shows a visualization of the spacecraft's magnetic field model.[11]

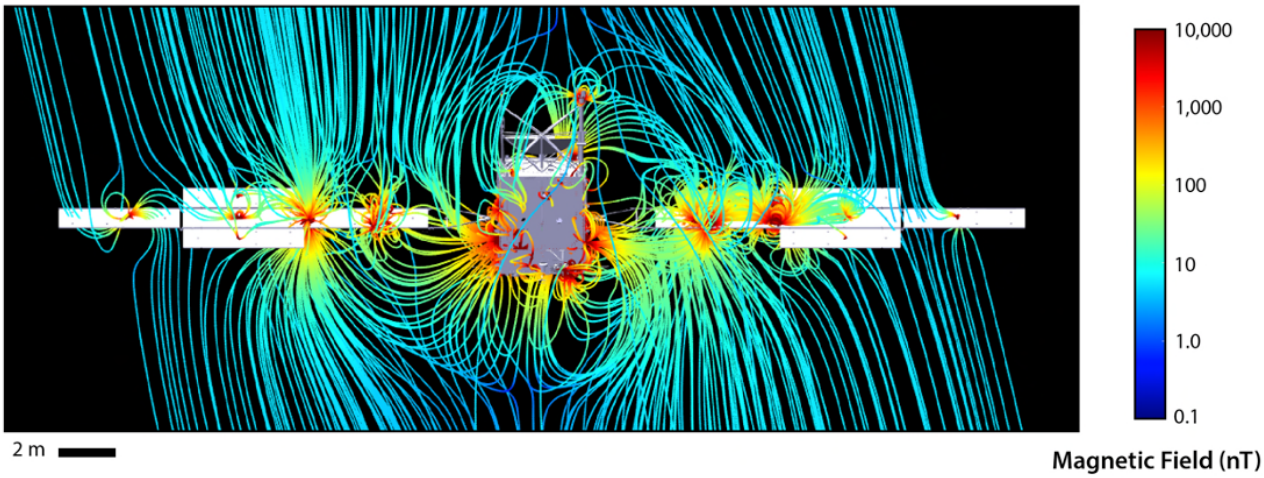


Figure 36. Psyche spacecraft magnetic field.
Magnetometers sensor units are at the middle and top of the boom on the top-left of the spacecraft's main body.[11,25]

The Magnetometer was able to observe EM signatures for the thrusters, cold gas desaturations, spacecraft heaters turning on and off, autonomous battery recharging, and environmental events such as solar coronal mass ejections (CMEs). Figure 37 shows an summary of the magnetic field per axis during ICO indicating observations of CMEs, short thrust events, and the week-long Long Duration System Test (LDST) thrusting checkout.

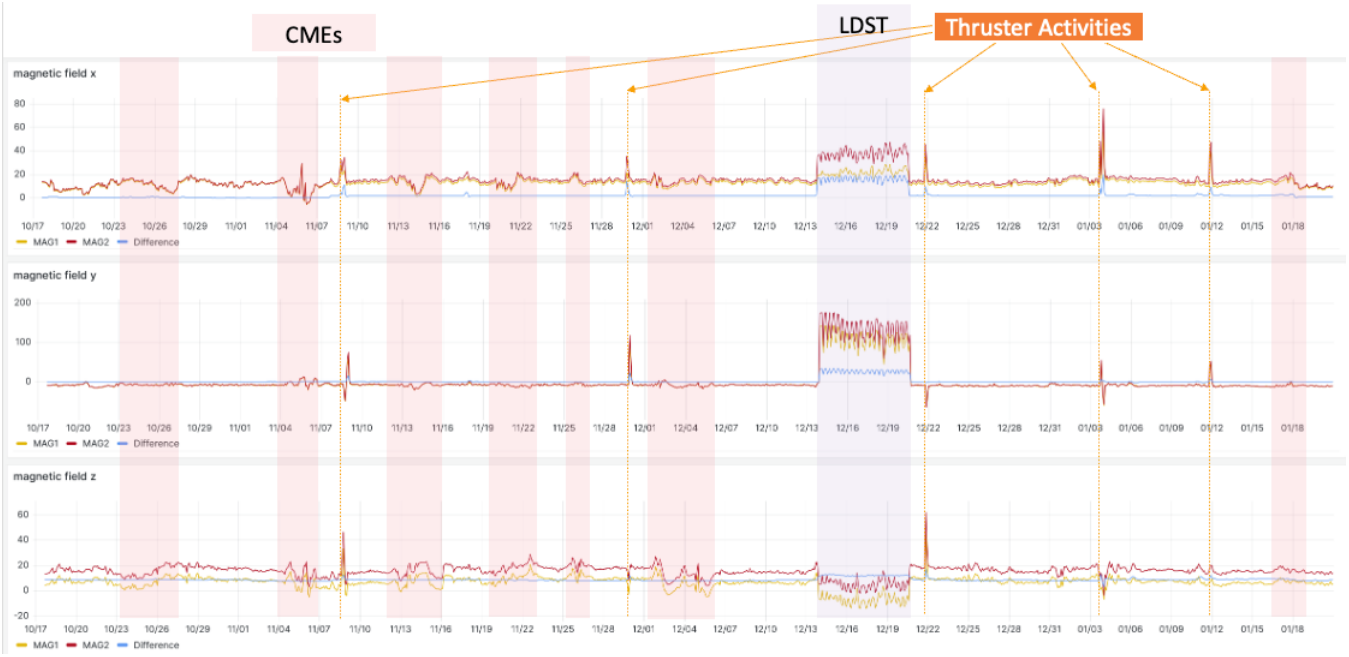


Figure 37. A summary of magnetic field measurements during ICO. Highlighted intervals show spacecraft thrusting activities and solar coronal mass ejections.

More data continue to be acquired as Psyche enters the continuous cruise thrusting phase. The team has found that the repeating nature of the EP thruster duty cycles and the relatively smooth step functions in the difference field may allow for the removal of the thruster field signatures in flight, as is shown in Figure 38. Originally it was hypothesized that magnetometer data while thrusting may be too noisy to be scientifically useful, however that may not be the case.

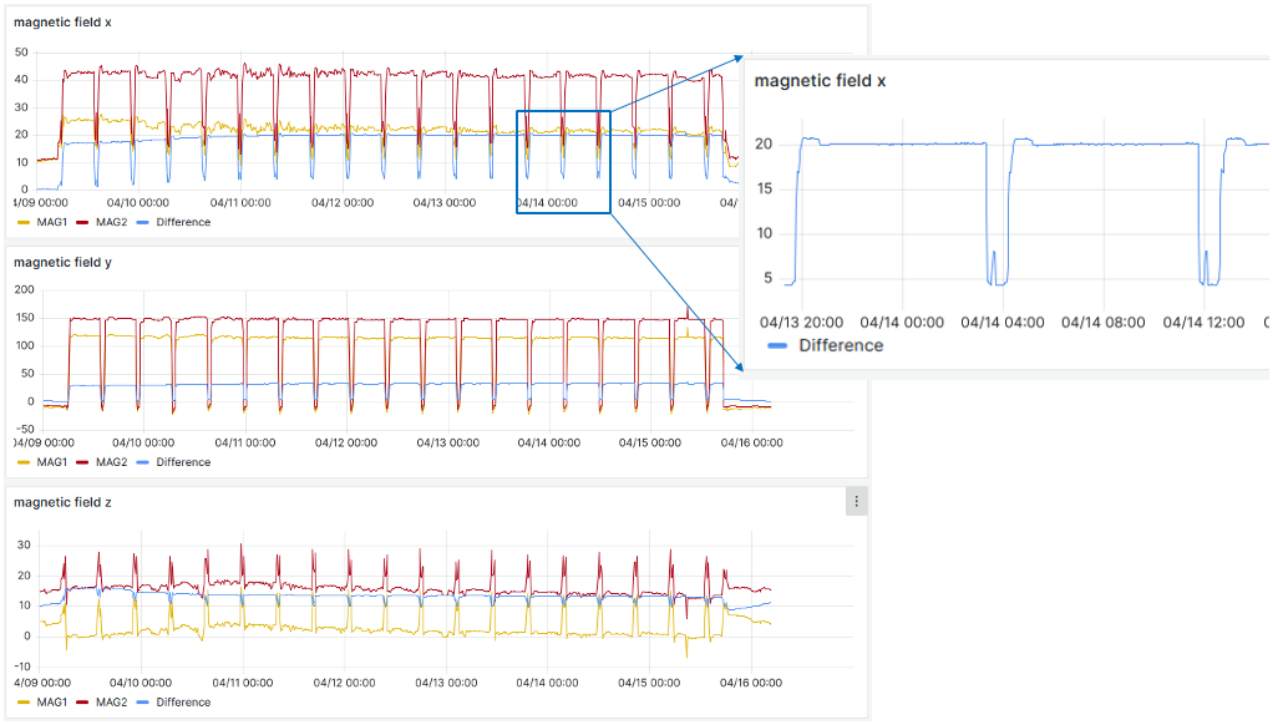


Figure 38. Magnetic measurements during thrusting.

The 50 Hz Magnetometer data have also been beneficial observing the known effect of EP thrusters stochastically

switching between two different stable plasma configurations during their burn in period. The Magnetometer initial indication of these events was that the magnetic field data would suddenly increase in data volume as the ability to compress the data decreased from nearly 100% to closer to 50% (see Figure 39). This indicated noise in the field and comparison to power processing unit (PPU) current telemetry during the associated timeframes shows a clear correlation. Additionally, the magnetic field data plots appear to resolve fast transitions that are missed at the resolution of the PPU data sampling rates. The Magnetometer team continues to collaborate with the Propulsion Team to characterize this EP thruster behavior.

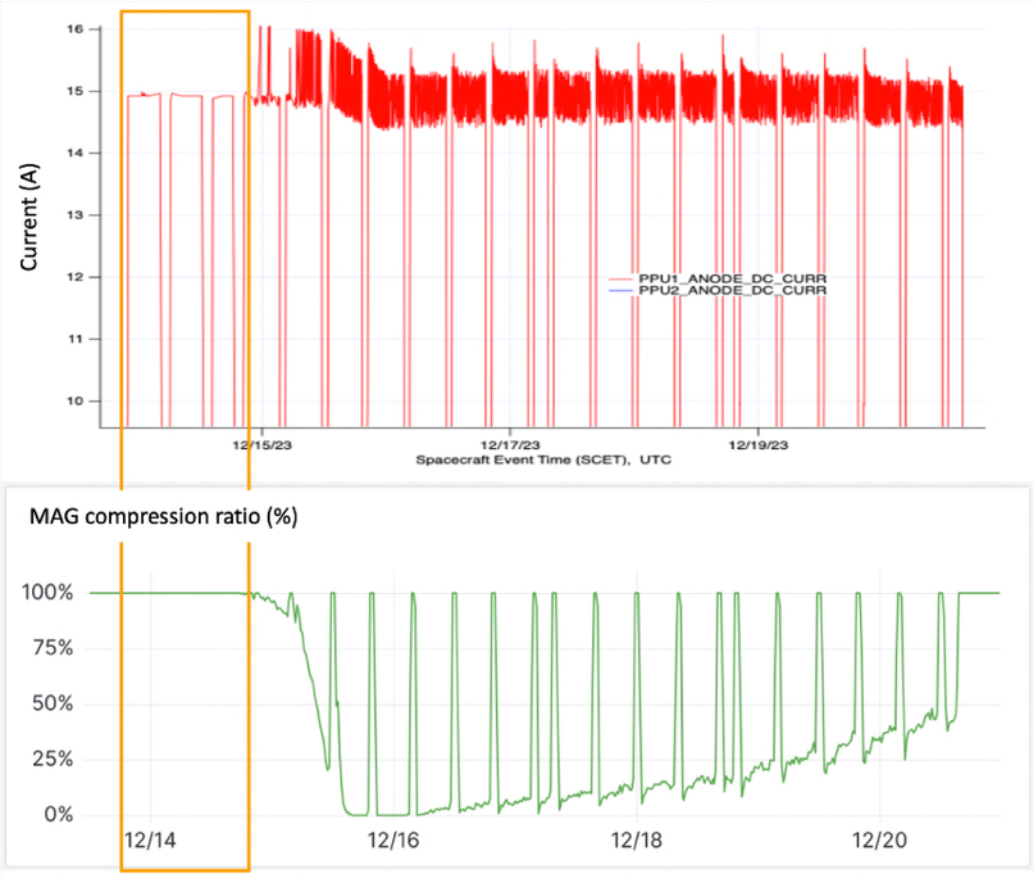


Figure 39. EP thruster anode current effects on Magnetometer data compression.
The orange box indicates the time period prior to the start of anode current oscillations.

The Psyche Multispectral Imager checkout campaign consisted of a bakeout activity, "first light" functional tests for both cameras, and more complex photometric and geometric star calibration observations. In November 2023, spacecraft heaters were utilized to raise Imager temperatures to ~60°C for 14 days to clear any residual moisture from the optics. A functional checkout in early December then provided the first images of the mission while exercising basic instrument functions such as power on/off, filter wheel cycling and imaging in all filters at varying exposures, and storing and transferring images from the Imager electronics to the spacecraft flight software for downlink. The same functional test was performed for both Imagers and took place at the nominal HGA-to-Earth downlink attitude as it required no specialized spacecraft pointing. These "first light" images proved that the cameras were operational and also confirmed there were no adverse impacts on camera performance from vibration and/or shock during the launch event.

In early January 2024, each Imager executed a ~9-hour photometric/radiometric and geometric star calibration activity. For these observations, specific spacecraft pointing was implemented to first target two different stellar targets: Canopus, a standard well-characterized star, and Kappa Ceti, a solar analog star. At each target, four exposures were acquired in each multispectral filter including two dark images with blocking/opaque filter, and one image with zero second exposure to characterize shutter smear. A 3x3 image mosaic centered on stellar cluster NGC869 was then performed to characterize geometric distortion effects. This activity included long and short exposures through each filter per mosaic position with two additional dark images with the blocking filter in place at the center mosaic position (see Imager field-of-view simulation in Figure 40; further testing/calibration details and actual flight images can be found in subsequent publications).[26,27]

The imaging ICO checkout campaign acquired a total of 250 images per camera. Data analysis and future calibration refinement will continue, but the team has assessed key characteristics and requirements for both Imagers such as: radiometric accuracy, detector readout noise, dark current and hot pixels, point spread function, and focal length. Images from the star calibrations have allowed the team to validate pre-flight measurements and show that the instruments are meeting requirements, with both systems behaving as expected with no detectable distortion in either Imager. These data and analyses will continue to feed into the development of the Imager's camera model including any geometric distortion. The team will have the opportunity twice a year to continue similar calibrations with the next star imaging activity during the mission's Periodic Maintenance and Calibration effort in July 2024.

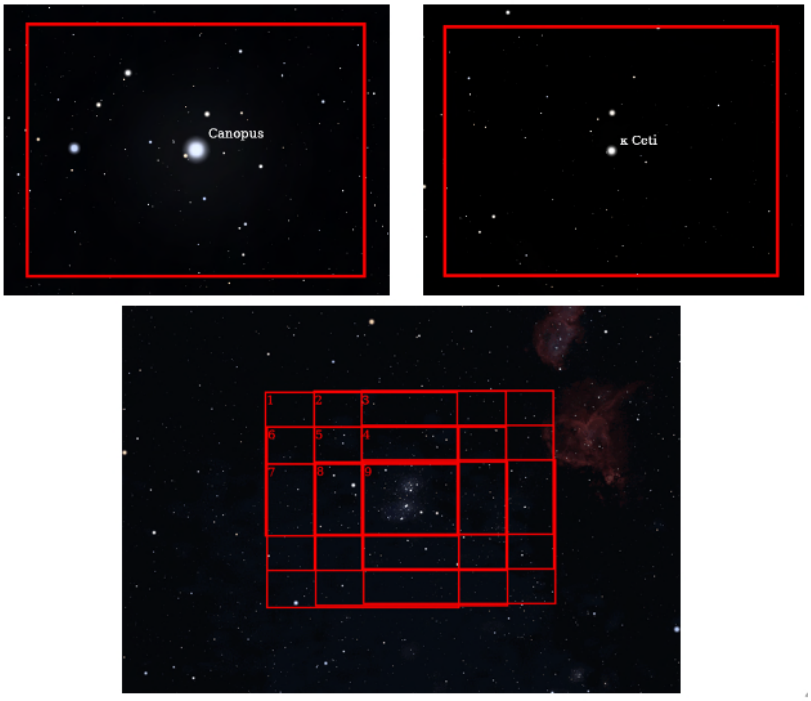


Figure 40. Psyche multispectral imager star calibration pointing design concepts.

The Gamma-Ray and Neutron Spectrometer investigation had separate commissioning efforts for each instrument. The Gamma-Ray Spectrometer (GRS) had the most involved series of checkout steps occurring over a 3-week period from November 6-27, 2023. The first step was to bake out the GRS cryostat for 48 hours to remove any contamination via outgassing. This also included a 2-hour cryocooler checkout to get an early assessment of performance before the full extended cooldown. High voltage was raised on the GRS Anti-Coincidence Shield (ACS) to inform environmental conditions and the team began assessing science performance and instrument parameter settings. On November 13, after 3 days of operating the cryocooler to cool the High Purity Germanium (HPGe) crystal to an acceptable temperature for full GRS checkout, operators carefully ramped up HPGe high voltage in incremental steps with ground verification of telemetry and instrument health along the way. The instrument then completed a two-week-long measurement of background gamma rays in the Psyche spacecraft environment with a couple of opportunities for instrument parameter tuning as well. This checkout activity early in the mission enabled data collection from the HPGe detector before significant radiation damage had occurred, which was critical to creating a baseline performance metric and informing the annealing strategy for the entire mission.

Results from the GRS commissioning are impressive. The first in-flight use of the Lockheed Martin cryocooler showed performance that exceeded expectations, maintaining the target 95 K temperature with less power than predicted (15 W). Measurement of the spacecraft background indicated the presence of some potassium, but virtually no nickel, a key element under investigation at the asteroid. This validated the pre-flight nickel-reduction efforts and implies GRS will meet its Ni measurement requirement with margin. Overall, spacecraft background signals are about one-third of previously seen heritage levels. The GRS HPGe performance is also outstanding with an inferred energy resolution of close to 2 keV at 1333 keV (see Figure 41), which is best in class for this type of gamma-ray spectrometer and exceeds any system previously flown in space.[28] In total, the team has detected more than 100 identifiable gamma-ray lines generated by various processes of galactic cosmic rays (GCRs) hitting instrument and spacecraft materials or the presence

of naturally radioactive materials. The ACS is also performing as expected and demonstrating its background rejection capability.

Commissioning of the Neutron Spectrometer (NS) took place on December 11, 2023. Like the GRS checkout, after initial power on, high voltage to the three ^{3}He neutron sensors was carefully ramped up to operative levels with ground-in-the-loop verification of performance as the activity progressed. NS collected high-quality spectral measurements, raw event-mode data, and identified spacecraft backgrounds. The data also informed later instrument parameter tuning. In-space data show higher background at low channels than was measured on Earth due to the galactic cosmic ray environment, though neutron peaks are still cleanly detected (see Figure 42).[28] Analysis of energy deposition spectra from each of the sensors indicate the dominant neutron energy range at the Psyche spacecraft is higher energy neutrons.

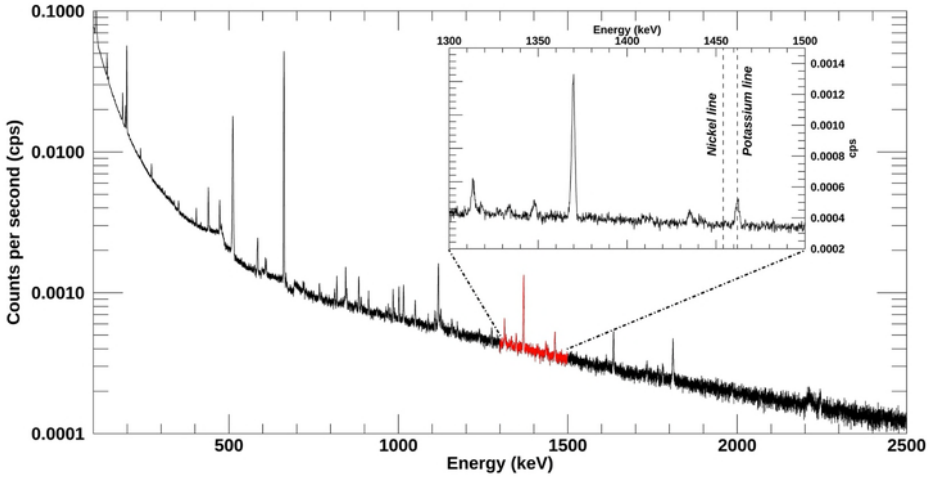


Figure 41. Gamma-ray spectrometer first measurements from space.

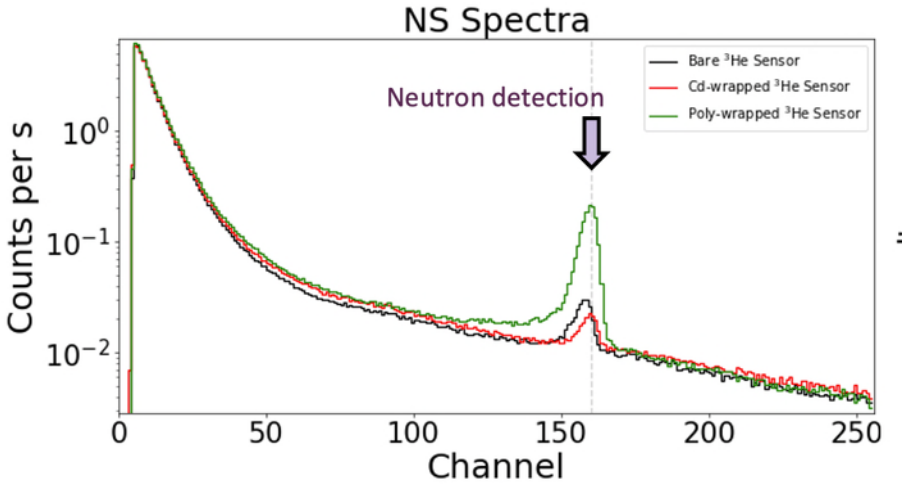


Figure 42. Neutron spectrometer spectra.

The GRS ACS and NS remain on and will collect space environment data throughout the journey to Psyche. This has allowed the investigation to use cruise data to act as an energetic particle "weather station" continually monitoring the charged particle environment (see Figure 43).[28] The ACS can detect galactic cosmic rays, and the three different NS sensor coverings provide varying energy thresholds for energetic proton detection. While close to Earth, the ACS count rates tracked Earth-based measures of solar energetic particles and GCR variability. Solar particle events (SPEs) detected by Psyche can be correlated with those detected by the Earth-orbiting GOES satellite.

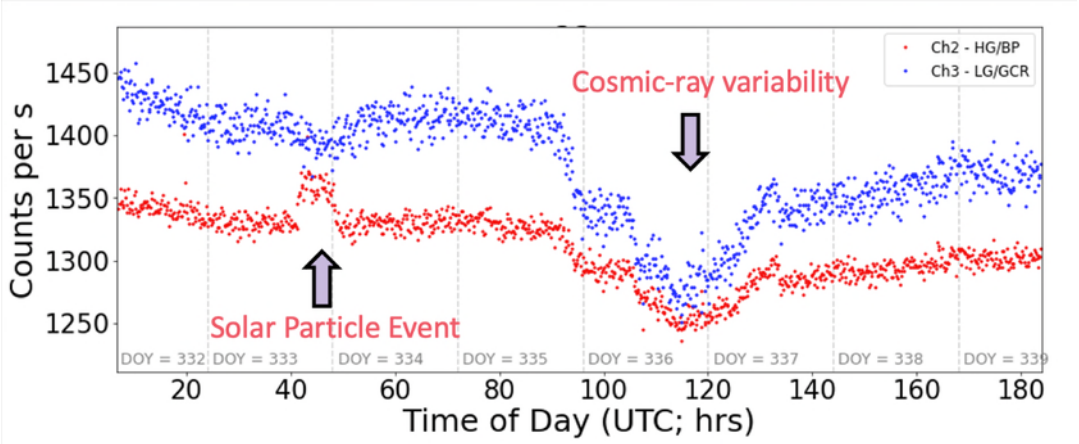


Figure 43. GRNS cruise data enable monitoring of the charged particle environment in space.

Overall, the ICO science instrument commissioning activities were highly successful. As the Magnetometer, GRS, and NS continue to collect data during the cruise thrusting phase of the mission, science teams are beginning to meet to analyze and collaborate with the results. Going forward, about every six months, an approximately 4 day pause in thrusting is planned to continue to checkout and calibrate all instruments during the planned Periodic Maintenance and Calibration opportunities.

G. DSOC Technology Demonstration

DSOC test activities were also designated as high priority, mandatory activities during ICO. The DSOC flight transceiver (FLT) onboard the Psyche spacecraft includes a 22-cm diameter off-axis optical transceiver assembly (OTA) for receiving a 1064 nm laser signal transmitted from Earth while returning a 1550 nm communications laser beam to the ground receiving station. A 4-watt average power laser transmitter along with electronics and a photon-counting camera (PCC) are included in the FLT. The FLT uses a mounting scheme that decouples the OTA from base mechanical vibrations, except for weak coupling through flexible cables needed for power and data transfer between the "stationary" and isolated or "floating" side. The actuators and sensors used for maintaining mechanical isolation allow steering of the isolated assembly over a limited field of regard, enabling fine pointing control within the accuracy uncertainty of the Psyche spacecraft's pointing capabilities. The OTA was locked in place for launch, and then released in space. Once released, the "floating" payload is actively controlled with actuators and sensors to maintain desired pointing. A docking mechanism arrests any OTA motion relative to the spacecraft when powered off, thereby preventing the uncontrolled payload from running into mechanical stops. Prior to every contact the payload is undocked and controlled with actuators and sensors to maintain a pose favoring bi-directional laser communications. The OTA downlink optical train includes a fast-steering mirror used to point the downlink "ahead" so that the laser will be incident on the ground station when it arrives at Earth, accounting for long light times and relative spacecraft transverse velocities. As described previously, the boresight of the FLT is aligned to the Psyche spacecraft HGA to allow the spacecraft to conduct simultaneous optical and radio communication with the Earth.

The DSOC ground subsystem includes a high power (~ 5 kW) ground laser transmitter (GLT) integrated to the existing Optical Communications Telescope Laboratory (OCTL) located near Wrightwood, CA. The link is initiated by transmitting the 1064 nm laser beam (uplink beacon) to the predicted Psyche spacecraft location in space. The Psyche spacecraft points the DSOC boresight towards Earth, allowing the FLT to acquire and track on the uplink beacon. The faint signal delivered to the DSOC FLT is sensed by the PCC and processed to enable its acquisition and tracking functions and to decode low rate data encoded onto the uplink laser. Pointing vectors generated onboard Psyche and streamed to the DSOC electronics are used to steer the downlink beam to the ground laser receiver (GLR), located at the Palomar Observatory, Hale telescope. A photon-counting receiver (PCR) consisting of 1) a superconducting nanowire single photon detector (SNSPD) and 2) a signal processing assembly are retrofitted to the Hale telescope. The downlink collected by the 5-m diameter telescope is detected by the SNSPD and the counts are processed to achieve decoding of information transmitted from deep space. DSOC adopts the Consultative Committee for Space Data Systems (CCSDS), High Photon-Efficiency (HPE) signaling standard and is the first instantiation of this standard. With this scheme more than one bit/photon downlink performance is achieved. Figure 44 shows photos of the DSOC system hardware described above.

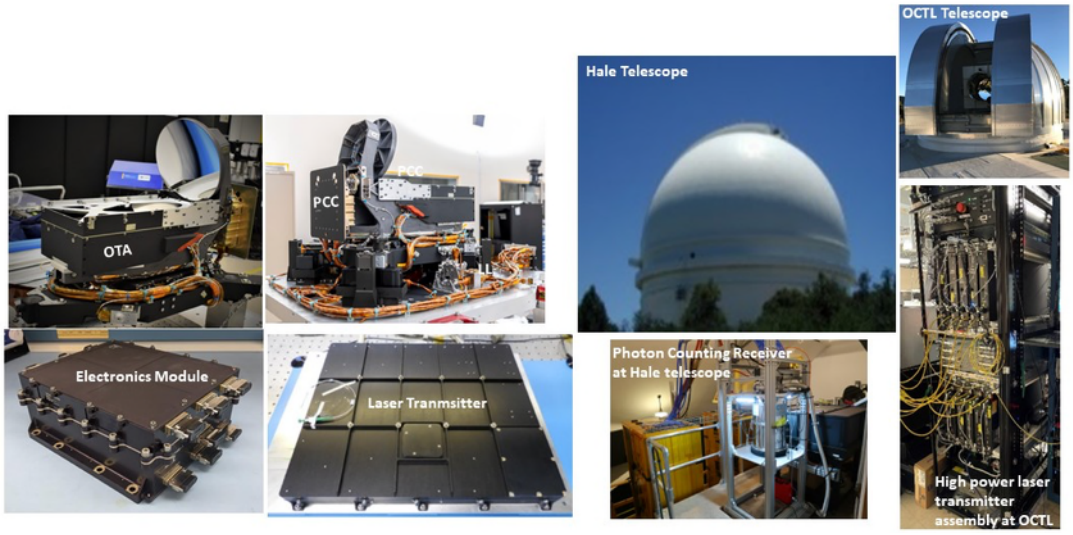


Figure 44. Showing the key assemblies of the DSOC flight transceiver and the DSOC ground stations.

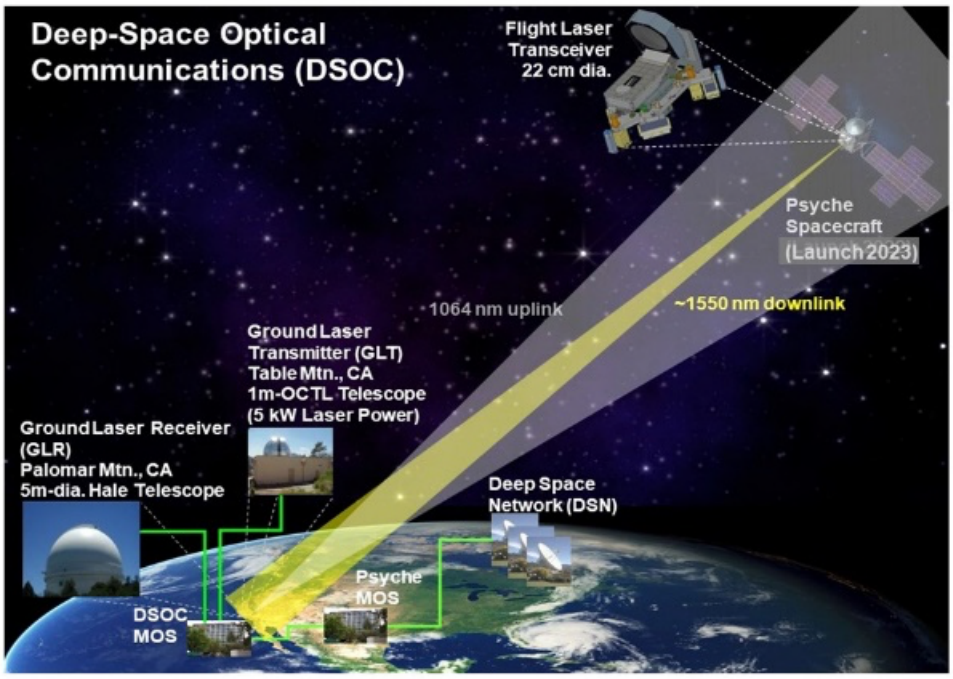


Figure 45. An operational view of the DSOC technology demonstration.

Figure 45 shows an operational view of the DSOC technology demonstration. A few weeks after launch the protective aperture cover was deployed, and launch-locks were released. First power-on was executed and mechanical disturbance data in the space environment was characterized. Telemetry received over the Psyche telecommunications link allowed the operations team to verify that the DSOC FLT was functioning nominally in space. DSOC commissioning required active participation of the Psyche operations team to approve and radiate FLT commands in real-time over the Psyche telecommunications link during the pass. The following checkouts were completed during commissioning: 1) correction of pointing offsets between Psyche derived pointing vectors and DSOC OTA boresight; 2) acquisition of the uplink beacon and low-rate data transfer at 1.8 kb/s 3) uplink tracking and optimizing downlink pointing and 4) establishing sustained bi-directional optical links at the designed maximum data-rate of 267 Mb/s. To demonstrate link utility, pre-loaded high-definition videos and imagery were transferred over the downlink. The optical uplink was used for limited commanding of the FLT to make data-rate changes or command small pointing corrections based on flight-ground pointing calibration scans, processed on the ground. DSOC's commissioning phase ended in December 2023.

DSOC operations started on January 2, 2024, and is continuing once a week. The entire contact period is operated with pre-loaded command sequences. Ground stations coordinate activities with the spacecraft schedule. No Psyche operator-in-the-loop is required during the DSOC passes. DSOC top level (Level 1) performance requirements were met and surpassed, with downlink data-rates decreasing consistently with the inverse square distance law. Link performance has exceeded the conservative pre-launch predictions by at least a factor of 2 while using half the available average laser power and modulation formats that restricted the full peak power. The laser was operated de-rated to extend its lifetime, with plans to implement full capability in the latter half of the demonstration. Figure 46 shows a summary plot of data rates achieved using the GLR at Palomar observatory and with the 1-m OCTL telescope, when the distances were closer to Earth. The single OCTL data point at 0.8 AU has the downlink pointed at OCTL instead of downlink being pointed at Palomar.

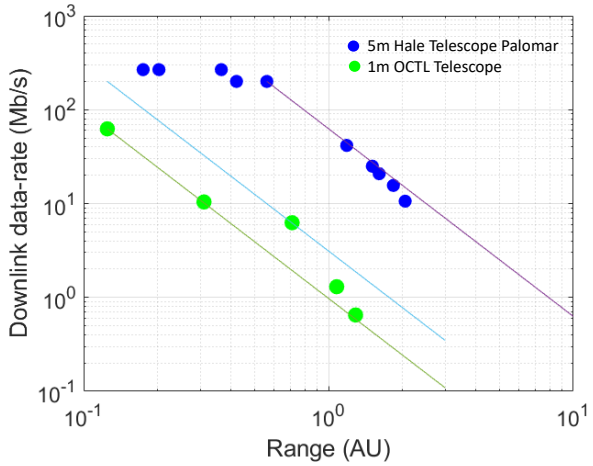


Figure 46. Summary of DSOC downlink data-rate versus distance, the lines represent inverse square distance. ***Palomar data rates deviating from the line are due to operating with higher link margin.***

The ability to return Psyche spacecraft engineering data and transfer high-rate DSOC telemetry has been demonstrated. For example, minutes of 60 Hz camera frames and 15 kHz sampled uplink power which could not be accommodated by the Psyche-allocated RF downlink data volume were made available for analysis via the optical downlink. A loop-back mode of sending image files on the uplink and returning them on the downlink was also demonstrated. As of mid-May 2024, the Psyche spacecraft was last contacted at 2.06 AU with a downlink data-rate of 10.4 Mb/s. This is well in excess of the 240 kbps maximum rate achievable from the same range using Psyche's X-band HGA.

DSOC plans to continue operations through the September 2025 with an interruption from July-Dec. 2024. The technology demonstration has achieved all the technical objectives, so far, and will serve as a significant milestone on NASA's deep space communications plan.

VI. Conclusion

This paper has provided an overview of the Psyche project's progression through launch and its first six months of operations in space, including its initial checkout (ICO) activities. After launch in October 2023, the spacecraft successfully separated from its Falcon Heavy launch vehicle, stabilized its attitude, executed its autonomous solar array deployment and established communications with the ground operations team. Then, over the next 103 days, the team executed the spacecraft's ICO plan and successfully verified nominal operation of the spacecraft. A major milestone was achieved with the successful execution of the Long Duration System Test, which validated the mission's unique all-electric momentum management strategy and effectively demonstrated the project's ability to execute sustained cruise thrusting with its Hall thruster based electric propulsion system – an essential first-time demonstration and enabling capability for the mission.

Psyche also completed commissioning and checkout of its Deep Space Optical Communication (DSOC) technology demonstration payload, including establishing sustained bi-directional optical links with the Earth from deep space at its designed maximum data-rate of 267 Mb/s. The technology demonstration has achieved all its technical objectives so far, and plans to continue operation through September 2025. Its success and will serve as a significant milestone on NASA's deep space communications plan.

While the spacecraft is operating nominally overall, based on the results of ICO, several issues have been identified for further investigation. These include higher than expected voltage on the solar arrays, out-of-family burn-in behavior

for EP Thruster B, and higher than anticipated measurements of swirl torque seen on all thrusters during extended calibration maneuvers. The cadence of EP momentum management maneuvers has been higher than anticipated, and the operations team is following multiple avenues of improvement to reduce this cadence going forward. The first few months of Psyche operations have shown that a turn-and-burn strategy for managing swirl momentum from Hall thrusters is effective, but operationally complex. Future missions should seek opportunities to reduce this operational complexity via hardware mitigations and/or changes to onboard GNC flight software algorithms which could greatly reduce or eliminate the time spent scheduling and conducting momentum management maneuvers. In particular, thruster designers could greatly simplify swirl torque momentum management by flight qualifying a PPU and Hall thruster with the capability to change the direction of the thruster's magnetic field prior to each startup.

Overall, Psyche's launch and ICO activities have been a success, the spacecraft has completed its transition to cruise operations, and has started cruise mission thrusting. The spacecraft has now successfully begun its 2.2-billion-mile journey to the largest metal-rich asteroid in the solar system, with its next major mission milestone being a Mars Gravity Assist scheduled to occur in May 2026.

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During the preparation of this work the authors used ChatGPT to do grammatical and text editing of parts of section III. After using this tool, the authors reviewed and edited the content and take full responsibility for the content of the publication.

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